

Slip-Cast Ceramic Ocarina (OCA-001) — v3 Smoke Test

Instrument-Maker Print Packet

Build packet folder: /sessions/charming-bold-fermat/mnt/GitHub/ocarina

Generated: 2026-05-02

This packet is the printable companion to the build folder. Take it shopping or into the shop. Tear sheets at page breaks.

File Map

File	Purpose
design.md	Project intent, catalog metadata, assumptions, and validation plan.
bom.csv	Starter bill of materials with part categories, quantities, drawing refs, and notes.
sourcing.csv	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
cut-list.csv	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
drawing-brief.md	Manufacturing drawing and technical product sketch brief.
assembly-manual.md	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
validation.csv	Target/measured values, tolerance, environment, result, and tuning/build action log.
supplier-rfq.md	Supplier email/request-for-quote starter.
visual-bom-brief.md	Art direction for an image-forward visual BOM.
wolfram-starter.wl	Wolfram starter for physics, optimization, visualization, and validation.
README.md	Project artifact.
capstone-deck.md	Project artifact.
print-packet.md	Project artifact.

design.md

Project intent, catalog metadata, assumptions, and validation plan.

Ocarina Build Packet

Source

- Design table: `docs/Ocarina.xlsx`
- Sheet: `Ocarina`
- Inspected range: `A1:H141`
- Workbook content observed: design inputs, size library, Helmholtz calculator, 12-hole fingering chart, ceramic workflow, BOM, design notes, and Wolfram notebook notes.

Design Intent

Build a slip-cast ceramic Alto C 12-hole ocarina using a 3D-printed master and plaster mold workflow. The current table is focused on the standard "sweet potato" vessel-flute form, but the same model can be used as the baseline for sculptural vessel flutes.

Governing Model

Ocarinas behave primarily as Helmholtz resonators:

$$f = c/(2\pi) * \sqrt{A_{\text{open}}/(V_{\text{chamber}} * L_{\text{eff}})}$$

Unlike a transverse flute or Native American style flute, pitch is driven by chamber volume and total open-hole area, not tone-hole distance along a bore. Hole position is still important for ergonomics, grip, and fingering logic.

Current Workbook Inputs

Field	Current value
---	---
Ocarina type	Alto C, 12-hole
Chamber volume	130 cm ³
Wall thickness	0.4 cm
Voicing window width	0.9 cm
Voicing window height	0.5 cm
Speed of sound	34300 cm/s
Clay body	Cone 6 stoneware

Shrinkage	12 percent
Target low note	A4, 440 Hz
Target high note	F6, 1397 Hz

Critical Design Features

- Chamber volume is the master tuning input. Measure it by water fill after CAD, master print, greenware, bisque, and glaze fire.
- Voicing window area sets the all-closed low-note behavior along with volume and effective neck length.
- Fipple geometry is the playability bottleneck: windway gap, labium angle, channel smoothness, and edge sharpness should be treated as first-class dimensions.
- Finger holes should start undersized. Enlarge to sharpen; flattening after over-enlarging is possible but slower and less clean.
- Exterior-only glazing is preferred for tuning stability and acoustic response. Mask the fipple, holes, and chamber interior.

Workbook Improvement Notes

During inspection, rows 39-42 of the 12-hole fingering chart contain formulas for hole area, cumulative open area, and calculated frequency. Rows 43-51 currently show placeholder `=` cells instead of filled formulas. Recommended next workbook cleanup:

1. Copy the area formula pattern from `D40:D42` through `D51`.
2. Copy the cumulative area pattern from `E40:E42` through `E51`.
3. Copy the frequency formula pattern from `F40:F42` through `F51`.
4. Add measured frequency, cents error, and tuning action columns.
5. Add a note that hole combinations can be cross-fingered if total open area is equivalent.
6. Verify the size-library `Length (in)` column. Inspection showed serial-like values such as `46180`, so those cells may have been formatted or entered incorrectly.

Mold Strategy

Use a 2-piece plaster mold for the main body. The first prototype can use a press-molded body to speed iteration, but production should move toward slip casting once the fipple and chamber volume are stable.

Recommended mold/CAD features:

- Split line around the widest body perimeter.
- Positive registration keys that do not trap clay.
- Pour/drain path or assembly opening appropriate to the chosen casting workflow.

- Removable or cleanable fipple detail. If the fipple is too delicate for the mold, cast a blank and hand-cut the voicing at leather-hard stage.
- Master scale factor: $\frac{1}{(1 - \text{shrinkage})}$; for 12 percent shrinkage this is about 1.136.

Prototype Ladder

| Prototype | Goal | Success criteria |

| --- | --- | --- |

| OCA-P0 voicing tile | Practice windway/labium only | Clear tone on a disposable test cavity |

| OCA-P1 closed vessel | Verify chamber volume and fipple | Stable A4-ish all-closed tone before finger holes |

| OCA-P2 4-hole | Prove tuning workflow | Four tuned notes within +/-25 cents after bisque |

| OCA-P3 12-hole | Full range test | Chromatic pattern playable, no blocked grip |

| OCA-P4 matched set | Soprano C, Alto C, Bass C | Shared mold logic and glaze family |

Open Assumptions

- Pricing in the workbook is an estimate and has not been date-checked.
- Clay shrinkage must be measured for Tony's actual slip, not assumed from supplier average.
- The fipple may need empirical iteration beyond the Helmholtz model.
- Exact ergonomics depend on hand span and final sculptural body shape.

bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item_id	category	item	qty	spec	make_buy	estimated_cost	source_note	drawing_ref	notes
OCA-BOM-001	Master	3D printed body master	1	PLA or resin print	Make	\$2-5	Workbook estimate	OCA-Draw-001	Scale master for measured clay
OCA-BOM-002	Mold	#1 pottery plaster	10 lb	USG #1 pottery plaster	Buy	\$15-25	Workbook estimate	OCA-Draw-002	Enough for several small molds
OCA-BOM-003	Clay	Cone 6 stoneware	25 lb	Slip or clay body	Buy	\$20-35	Workbook estimate	OCA-Draw-003	Record batch and measured slip
OCA-BOM-004	Finish	Cone 6 exterior glaze	1.5 pints	Food-safe or durable	Buy	\$30-60	Workbook estimate	OCA-Draw-004	Mask fipple, holes, and interior
OCA-BOM-005	Tuning	Diamond needle file	1 set	Assorted profiles	Buy	\$10-20	Workbook estimate	OCA-Draw-005	Needed after bisque and final
OCA-BOM-006	Tuning	Chromatic tuner	1	or analysis app	Buy	\$15-30	Workbook estimate	OCA-Draw-006	Use same app/settings across
OCA-BOM-007	Mold	Mold soap or release	1 bottle	Murphy's oil soap	Buy	\$5-10	Workbook estimate	OCA-Draw-007	For sealed master and plaster
OCA-BOM-008	Firing	Bisque and glaze firing	1 load	Cone 06 bisque and glaze	Buy	\$20-50/load	Workbook estimate	OCA-Draw-008	Record firing schedule and kiln
OCA-BOM-009	Masking	Wax resist	1 small jar	Brushable wax resist	Buy	TBD	Add supplier/date	OCA-Draw-009	Protect tuning-critical holes and
OCA-BOM-010	Measurement	Graduated syringe or burette	1	Water-fill volume	Buy	TBD	Add supplier/date	OCA-Draw-010	Needed for chamber volume v

sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

item_id	item	required_spec	search_terms	supplier_candidate	date_checked	unit_price	lead_time	substitution_risk	note
OCA-SRC-001	#1 pottery plaster	Pottery mold plaster	US Gypsum plaster	10 lb bags				Use equivalent plaster	Very brittle if too dry or overkneaded
OCA-SRC-002	Cone 6 casting slip	Stoneware slip with iron oxide stain	10 lb bags					Substitute only after news using bagging labels	Can damage mold
OCA-SRC-003	PLA or resin	Print material that can be used in a mold making resin print	PLA filament	1 lb spools				Any stable printable filament	Can damage mold
OCA-SRC-004	Mold release	Mold soap compound	Water-based mold release					Commercial mold release	Can damage mold
OCA-SRC-005	Diamond files	Small round and tapered diamond files	1/8" and 1/16" files					Diamond burrs or burs	Can damage mold
OCA-SRC-006	Wax resist	Brushable wax for glaze masking	1 lb jar					Latex or tape only	Can damage mold
OCA-SRC-007	Measurement tool	Cent-accurate tool for measuring volume	100 ml graduated syringe					Phone app accepted	Can damage mold

cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

cut_id	part	qty	rough_dim	final_dim	material	orientation	operation	tolerance_in	yield_or_offcut	notes
OCA-CUT-001	Master body upper half (3D-printed)	1	4.50 x 3.50 x 2.50	4.50 x 3.50 x 2.50	PEEK	Random	Injection molding	±0.05	100%	Part composite
OCA-CUT-002	Master body lower half (3D-printed)	1	4.50 x 3.50 x 2.50	4.50 x 3.50 x 2.50	PEEK	Random	Injection molding	±0.05	100%	Part composite
OCA-CUT-003	Wooden cottle boards (mold making)	8	1.00 x 0.75 x 0.25	1.00 x 0.75 x 0.25	Maple	Grain parallel	CNC machining	±0.01	100%	Use for mold
OCA-CUT-004	Plaster mother mold half (pourable)	2	5.00 x 5.50 x 2.00	5.00 x 5.50 x 2.00	Plaster	Random	Hand poured	±0.10	100%	Use for casting
OCA-CUT-005	Greenware vessel (slip-cast)	1	~3.95 x 3.07 x 2.20	~3.95 x 3.07 x 2.20	Greenware	Random	Slip casting	±0.05	100%	Part composite
OCA-CUT-006	Voicing window cut (leather-hard)	1	N/A	0.354 x 0.197	Wood	Grain parallel	CNC machining	±0.001	100%	Part composite
OCA-CUT-007	Finger holes (12 hole layout)	12	N/A	0.16-0.31 dia	Wood	Grain parallel	CNC machining	±0.001	100%	Part composite
OCA-CUT-008	Post-bisque tuning pegs	12	N/A	Within +/-25 cent	Wood	Grain parallel	CNC machining	±0.001	100%	Part composite
OCA-CUT-009	Wax resist mask (application)	1	N/A	~0.020 in dry film	Wax	Random	Hand applied	±0.001	100%	Part composite
OCA-CUT-010	Glaze application (application only)	1	N/A	~0.005-0.010 in	Glaze	Random	Hand applied	±0.001	100%	Part composite

drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Ocarina Drawing Brief

Required Views

- Top view with body outline, hole layout, hand assignment, and centerline.
- Side view with mouthpiece axis, windway height, labium angle, body thickness, and split line.
- Section through fipple showing windway, labium, window, and chamber connection.
- Section through body showing chamber volume reference, target wall thickness, and mold split.
- Detail view for hole diameters and post-fire tuning allowance.
- Mold view showing plaster parting line, registration keys, pour/drain plan, and cottle clearance.

Critical Dimensions

Dimension Source Tolerance intent
--- --- ---
Chamber volume Workbook input and water-fill measurement Tuning critical
Voicing window width/height Workbook input Playability critical
Windway gap CAD/test-piece measurement Playability critical
Labium angle and edge thickness CAD/test-piece measurement Playability critical
Wall thickness Workbook input and cast measurement Tone/drying critical
Hole diameters Workbook chart and tuning log Tuning critical
Master scale factor Measured clay shrinkage Process critical

Notes For CAD

- Store chamber volume as a named parameter.
- Store shrinkage factor as a named parameter.
- Separate final fired dimensions from master dimensions.
- Mark fipple geometry as prototype-sensitive; do not freeze it until test pieces speak reliably.
- Include a versioned body ID so castings can be tied back to the CAD master.

assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Ocarina Assembly Manual

Scope

This manual covers a slip-cast or press-molded ceramic ocarina based on `docs/Ocarina.xlsx`. It is written for prototype builds where the first goal is repeatable tone and validated tuning, not decorative finish.

Tools

- CAD package for master design.
- 3D printer and sanding/sealing supplies.
- Cottle boards, plaster mixing bucket, scale, mold soap, and clamps.
- Stoneware slip or workable clay body.
- Small hole cutters, drill bits, needle tools, loop tools, ribs, sponge, and fettling knife.
- Chromatic tuner, microphone, calipers, graduated syringe or water-fill vessel.
- Diamond needle files or fine diamond burrs.
- Wax resist, glaze, and kiln access.

Process

1. **Set design inputs**
 - Confirm target ocarina size, chamber volume, voicing window, wall thickness, clay body, and shrinkage.
 - Create a build ID before CAD begins.
2. **Model the master**
 - Model the body as a controlled-volume vessel.
 - Include split-line planning, registration features, and a fipple strategy.
 - Scale the master by $\frac{1}{1 - \text{measured_shrinkage}}$.
3. **Print and finish the master**
 - Print the master with enough wall strength to survive mold making.
 - Sand layer lines and seal the surface.
 - Mark datums: centerline, mouthpiece axis, split plane, and hole reference side.
4. **Make the mold**

- Apply release to the sealed master and cottle surfaces.
- Pour plaster with at least 1.5 inches around the master where practical.
- Add registration keys and dry the mold fully before casting.

5. ****Cast or press the body****

- For slip casting, pour slip, wait for target wall buildup, drain, and demold at leather-hard stage.
- For press molding, press even clay slabs into both mold halves and join with scored slip.
- Measure sample wall thickness at a noncritical trimmed area.

6. ****Cut fipple and holes****

- Establish the windway first.
- Cut the labium cleanly and test blow before the body fully dries.
- Cut finger holes undersized.

7. ****Dry slowly****

- Dry under plastic until moisture equalizes.
- Watch the mouthpiece and seam areas for cracks.

8. ****Bisque fire****

- Bisque to the clay body's recommended schedule.
- Measure mass, volume, hole diameters, and frequencies.

9. ****Tune after bisque****

- Enlarge holes to raise pitch.
- Refine the fipple only in small steps.
- Record measured frequency and cents error after each change.

10. ****Glaze****

- Wax resist the windway, labium, holes, and any interior opening.
- Keep glaze away from the acoustic edges.

11. ****Final fire and validation****

- Fire to the chosen glaze schedule.
- Re-measure all tuning points.
- Save final dimensions and tuning data back into the validation log.

Failure Modes To Watch

- Weak or no tone: windway too tall, labium too blunt, air jet missing edge.
- Pitch too flat: chamber too large, open area too small, or neck length too long.

- Pitch too sharp: chamber too small, open area too large, or over-filed holes.
- Warped mouthpiece: uneven drying or thin unsupported fipple geometry.
- Glaze-choked note: glaze reduced hole or windway opening.

validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

build_id	stage	date	clay_body	linkage	gas_steps	scale	lip_size	throat	throat_wiring	Hole	id	target	target	measured	tolerance	result	notes
OCA-P0	voicing_tile									all_closed	A4	440					Practice fipple before firing
OCA-P1	greenware					130	0.4	0.9	0.5	all_closed	A4	440					Measure before drying
OCA-P1	bisque					TBD	TBD	TBD	TBD	all_closed	A4	440					First real model check.
OCA-P1	glaze_fire					TBD	TBD	TBD	TBD	all_closed	A4	440					Record glaze shift.
OCA-P2	bisque					TBD	TBD	TBD	TBD	hole_1	Bb4	466.2					Start undersized and e
OCA-P2	bisque					TBD	TBD	TBD	TBD	hole_2	B4	493.9					Record hole diameter a
OCA-P2	bisque					TBD	TBD	TBD	TBD	hole_3	C5	523.3					Check breath pressure
OCA-P3	glaze_fire					TBD	TBD	TBD	TBD	hole_12	A5	880					Full 12-hole validation

supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ — Slip-Cast Ceramic Ocarina (OCA-001 family)

> Use this as a template for outreach to ceramic-supply, plaster, mold-making, and finishing suppliers. Customize the Subject and address fields per recipient.

Subject: RFQ — slip-cast ceramic ocarina prototype materials and consumables

Hello,

I'm a mechanical R&D engineer prototyping a small family of slip-cast ceramic ocarinas (vessel flutes, ~3-4 in long, ~0.16 in / 4 mm wall) and need quotes for the following. The first build is an Alto C 12-hole prototype based on the parametric Helmholtz design table at [tonykoop/ocarina](https://github.com/tonykoop/ocarina).

Items

| # | Item | Required spec | Approx. qty (per prototype run) |

|---|---|---|---|

| 1 | Cone 6 stoneware casting slip | Mature Cone 6 stoneware slip with **published shrinkage** (X/Y/Z), **water absorption %**, and **firing schedule**. White or light buff body preferred for tone consistency over dark iron-bearing slips. | 1-5 gal |

| 2 | #1 pottery plaster | USG #1 Pottery Plaster (or equivalent) suitable for absorbent slip-cast molds. Need **water:plaster ratio**, **set time**, **soluble salt content**. | 10 lb (single mold) — 50 lb (production mold set) |

| 3 | Mold release / mold soap | Plaster-compatible parting agent (Murphy's-type oil soap or commercial mold soap). | 1 bottle / 16 oz |

| 4 | Cone 6 exterior glaze | Food-safe (or durable non-food-safe) satin or matte glaze. Color: TBD per family — likely earth/buff/cream. Glaze must not bridge holes or windway. | 3-5 pints |

| 5 | Brushable wax resist | Standard ceramic wax resist for masking voicing window, finger holes, and chamber interior pre-glaze. | 1 small jar / 8 oz |

| 6 | Diamond needle file set (post-bisque tuning) | Assorted profiles (round, half-round, taper) ~3-6 mm; resin-bonded acceptable. | 1 set |

| 7 | Hole cutters / brass tube cutters | Round cutters from ~0.156 in (4 mm) to ~0.313 in (8 mm) diameter for greenware finger-hole cutting. | 1 set |

| 8 | Bisque + glaze firing service (if local) | Cone 06 bisque + Cone 6 glaze fire; small cubic-foot kiln load. | 2 fires per prototype |

| 9 | (Optional) Master printing/outsource | If you do high-resolution PLA or resin prints up to ~5 in long with **0.005 in dimensional accuracy**, please quote. We currently print in-house but want a backup. | 1-5 master pairs |

What we need in your quote

- **Unit price** and any volume breaks
- **Minimum order quantity**
- **Current stock status** and **lead time**
- **Shipping estimate** to 94566 (Pleasanton, CA) — or local pickup option
- **Material safety data sheet (MSDS)** and **technical data sheet** for items 1, 2, 4, 5
- **Recommended substitutions** for thin-walled (~4 mm) musical instrument bodies
- For item 1: most recent **shrinkage test data** for your batch (we will re-measure on our side, but a baseline matters)

Acceptance criteria

The prototypes are measured for X/Y/Z shrinkage, bore/cavity volume after firing, wall-thickness consistency, and acoustic tuning (chromatic tuner, ± 25 cents target post-bisque). Slip with unknown or wildly variable shrinkage invalidates the master scale factor (currently $1/(1-0.12) = 1.136$), so reproducibility data is more valuable than headline price.

The repo is a private R&D portfolio repo; we do not need any IP-restricted documents. Public-domain MSDS and supplier spec sheets are sufficient.

Thank you,

Tony Koop

Mechanical R&D Engineer

tonykoop@gmail.com

github.com/tonykoop/ocarina (private; access on request)

visual-bom-brief.md

Art direction for an image-forward visual BOM.

Ocarina Visual BOM Brief

Goal

Create a one-page visual BOM that helps a builder understand the ceramic ocarina workflow from digital master to tuned fired instrument.

Layout

- Header: "Slip-Cast Ceramic Ocarina - Alto C Prototype".
- Hero image: finished ocarina or clean CAD render.
- Process strip: CAD master, 3D print, plaster mold, greenware, bisque, tuning, glaze fire.
- BOM table: item number, part/material, quantity, spec, estimated cost, make/buy, image.
- Callout inset: fipple section with windway, labium, and voicing window labels.
- Validation inset: chamber volume, all-closed note, highest tested note, cents error.

Image Requirements

Use real shop photos when available. Generated or rendered placeholders should be labeled as placeholders until replaced with:

- Master print photo.
- Plaster mold photo.
- Greenware body photo.
- Fipple close-up.
- Final fired instrument.
- Tuning tool/photo of measurement setup.

wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

(* Ocarina Helmholtz notebook starter *)

```
ClearAll["Global`*"];
```

```
c = 34300; (* cm/s at about 20 C *)
```

```
targetA4 = 440;
```

```
volumeCm3 = 130;
```

```
wallCm = 0.4;
```

```
windowWcm = 0.9;
```

```
windowHcm = 0.5;
```

```
voicingAreaCm2[w_, h_] := w*h;
```

```
endCorrectionCm[area_] := 0.6*Sqrt[area/Pi];
```

```
helmholtzHz[area_, volume_, neckLength_] :=
```

```
(c/(2*Pi))*Sqrt[area/(volume*neckLength)];
```

```
centsError[measured_, target_] := 1200*Log[2, measured/target];
```

```
baseArea = voicingAreaCm2>windowWcm, windowHcm];
```

```
baseNeck = wallCm + endCorrectionCm[baseArea];
```

```
baseFreq = helmholtzHz[baseArea, volumeCm3, baseNeck];
```

```
holeDiameterMmToAreaCm2[d_] := Pi*(d/20)^2;
```

```
targetFreq[ midi_] := 440*2^((midi - 69)/12);
```

```
areaForFrequency[f_, volume_, neckLength_] :=
```

```
volume*neckLength*(2*Pi*f/c)^2;
```

(* Candidate cumulative area table for A4 to A5. *)

```
notes = {
```

```
{ "A4", 69}, {"Bb4", 70}, {"B4", 71}, {"C5", 72},
```

```
{ "C#5", 73}, {"D5", 74}, {"D#5", 75}, {"E5", 76},
```

```
{ "F5", 77}, {"F#5", 78}, {"G5", 79}, {"G#5", 80}, {"A5", 81}
```

```
};
```

```
areaTable = Table[
```

```
With[{freq = targetFreq[ midi]},
```

```
{name, freq, areaForFrequency[freq, volumeCm3, baseNeck]}
```

```
],  
{ {name, midi}, notes}  
];  
  
areaTable // TableForm
```

README.md

Project artifact.

Ocarina — Slip-Cast Ceramic Vessel-Flute Family

> *Engineering documentation for a 3D-printed-master, plaster-mold, cone-6-stoneware slip-cast ocarina family — from Helmholtz physics through the parametric design table to a manufacturable build packet.*

![[Hero — AI-generated concept render of a slip-cast Alto C 12-hole ocarina (placeholder until first prototype is photographed)]](images/hero-concept.

capstone-deck.md

Project artifact.

Slip-Cast Ceramic Ocarina (OCA-001) — v3 Smoke Test

- Musical instrument documentation capstone
- Build packet: ocarina
- Generated: 2026-05-02

Project Intent

- Build a slip-cast ceramic Alto C 12-hole ocarina using a 3D-printed master and plaster mold workflow. The current table is focused on the standard "sweet potato" vessel-flute form, but the same model can be used as the baseline for sculptural vessel flutes.

Speaker notes: Read design.md before committing to dimensions or sourcing decisions.

Physics Model

- Ocarinas behave primarily as Helmholtz resonators:

$$f = c / (2 * \pi) * \sqrt{A_{\text{open}} / (V_{\text{chamber}} * L_{\text{eff}})}$$

Speaker notes: Governing equation extracted verbatim from design.md. Apply empirical corrections (NAF K2, scale offsets) only where the model permits — see references/acoustic-models.md.

How To Use This Packet

- Start with design.md for intent and assumptions.
- Use bom.csv, sourcing.csv, and cut-list.csv before buying or cutting.
- Use drawing-brief.md and CAD/CNC folders before machining.
- Print the packet for shopping, shop work, and validation.

File Map

- design.md: Project intent, catalog metadata, assumptions, and validation plan.
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- supplier-rfq.md: Supplier email/request-for-quote starter.

Build Workflow

- Design and assumptions
- Source materials and hardware
- Prepare stock, fixtures, and CNC/laser/lathe setup
- Assemble, tune, finish, and validate

Sourcing And BOM

- BOM gives part categories and drawing references.
- Sourcing tracks search terms, supplier candidates, price/date, lead time, substitutions.
- Visual BOM brief turns the parts list into a presentation-ready image board.

Shop Packet

- Cut list for lumber/sheet/blank planning.
- Assembly manual for away-from-keyboard work.
- Validation sheet for measured dimensions, tuning, pass/fail checks.

Drawings, CAD, CNC

- drawing-brief.md defines required views, dimensions, datums, sketch intent.
- cad/ holds models and design tables.
- cnc/ holds CAM, toolpaths, setup sheets, dry-run notes.
- drawings/ holds PDFs, SVGs, DXFs, drawing exports.

![drawing](drawings/ocarina-body-section.svg)

![drawing](drawings/ocarina-hole-layout.svg)

![drawing](drawings/ocarina-mold-schematic.svg)

Images And Screenshots

- images/exploded-view-concept.png
- images/hero-concept.png
- images/mold-cross-section-concept.png

![[image]](images/exploded-view-concept.png)

![[image]](images/hero-concept.png)

![[image]](images/mold-cross-section-concept.png)

Validation Plan

- A4 = 440 Hz reference check.
- Tuning targets logged in validation.csv.
- Critical dimensions verified against design sheet and CAD.
- Photos and revision notes after each major step.

Open Risks / Decisions

- TBDs in design sheet and BOM.
- Supplier price/availability not yet verified.
- Generated images marked as concept placeholders.
- Empirical corrections await measured prototype data.

Next Actions

- Replace TBDs with measured/source-backed values.
- Verify live supplier price and availability before buying.
- Export final drawings and visual BOM images.
- Regenerate this deck and print packet after final edits.

print-packet.md

Project artifact.

Slip-Cast Ceramic Ocarina (OCA-001) — v3 Smoke Test Print Packet

Generated: 2026-05-02

Packet folder: `/sessions/charming-bold-fermat/mnt/GitHub/ocarina`

File Map

- | File | Purpose |
- | --- | --- |
- | `design.md` | Project intent, catalog metadata, assumptions, and validation plan. |
- | `bom.csv` | Starter bill of materials with part categories, quantities, drawing refs, and notes. |
- | `sourcing.csv` | Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks. |
- | `cut-list.csv` | Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts. |
- | `drawing-brief.md` | Manufacturing drawing and technical product sketch brief. |
- | `assembly-manual.md` | Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes. |
- | `validation.csv` | Target/measured values, tolerance, environment, result, and tuning/build action log. |
- | `supplier-rfq.md` | Supplier email/request-for-quote starter. |
- | `visual-bom-brief.md` | Art direction for an image-forward visual BOM. |
- | `wolfram-starter.wl` | Wolfram starter for physics, optimization, visualization, and validation. |
- | `README.md` | Project artifact. |
- | `capstone-deck.md` | Project artifact. |
- | `print-packet.md` | Project artifact. |

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design.md

Project intent, catalog metadata, assumptions, and validation plan.

Ocarina Build Packet

Source

- Design table: `docs/Ocarina.xlsx`
- Sheet: `Ocarina`
- Inspected range: `A1:H141`
- Workbook content observed: design inputs, size library, Helmholtz calculator, 12-hole fingering chart, ceramic workflow, BOM, design notes, and Wolfram notebook notes.

Design Intent

Build a slip-cast ceramic Alto C 12-hole ocarina using a 3D-printed master and plaster mold workflow. The current table is focused on the standard "sweet potato" vessel-flute form, but the same model can be used as the baseline for sculptural vessel flutes.

Governing Model

Ocarinas behave primarily as Helmholtz resonators:

$$f = c/(2\pi) * \sqrt{A_{\text{open}}/(V_{\text{chamber}} * L_{\text{eff}})}$$

Unlike a transverse flute or Native American style flute, pitch is driven by chamber volume and total open-hole area, not tone-hole distance along a bore. Hole position is still important for ergonomics, grip, and fingering logic.

Current Workbook Inputs

Field	Current value
---	---
Ocarina type	Alto C, 12-hole
Chamber volume	130 cm ³
Wall thickness	0.4 cm
Voicing window width	0.9 cm
Voicing window height	0.5 cm
Speed of sound	34300 cm/s
Clay body	Cone 6 stoneware
Shrinkage	12 percent
Target low note	A4, 440 Hz
Target high note	F6, 1397 Hz

Critical Design Features

- Chamber volume is the master tuning input. Measure it by water fill after CAD, master print, greenware, bisque, and glaze fire.

- Voicing window area sets the all-closed low-note behavior along with volume and effective neck length.
- Fipple geometry is the playability bottleneck: windway gap, labium angle, channel smoothness, and edge sharpness should be treated as first-class dimensions.
- Finger holes should start undersized. Enlarge to sharpen; flattening after over-enlarging is possible but slower and less clean.
- Exterior-only glazing is preferred for tuning stability and acoustic response. Mask the fipple, holes, and chamber interior.

Workbook Improvement Notes

During inspection, rows 39-42 of the 12-hole fingering chart contain formulas for hole area, cumulative open area, and calculated frequency. Rows 43-51 currently show placeholder `=` cells instead of filled formulas. Recommended next workbook cleanup:

1. Copy the area formula pattern from `D40:D42` through `D51`.
2. Copy the cumulative area pattern from `E40:E42` through `E51`.
3. Copy the frequency formula pattern from `F40:F42` through `F51`.
4. Add measured frequency, cents error, and tuning action columns.
5. Add a note that hole combinations can be cross-fingered if total open area is equivalent.
6. Verify the size-library `Length (in)` column. Inspection showed serial-like values such as `46180`, so those cells may have been formatted or entered incorrectly.

Mold Strategy

Use a 2-piece plaster mold for the main body. The first prototype can use a press-molded body to speed iteration, but production should move toward slip casting once the fipple and chamber volume are stable.

Recommended mold/CAD features:

- Split line around the widest body perimeter.
- Positive registration keys that do not trap clay.
- Pour/drain path or assembly opening appropriate to the chosen casting workflow.
- Removable or cleanable fipple detail. If the fipple is too delicate for the mold, cast a blank and hand-cut the voicing at leather-hard stage.
- Master scale factor: $1/(1 - \text{shrinkage})$; for 12 percent shrinkage this is about `1.136`.

Prototype Ladder

| Prototype | Goal | Success criteria |

| --- | --- | --- |

| OCA-P0 voicing tile | Practice windway/labium only | Clear tone on a disposable test cavity |

OCA-P1 closed vessel	Verify chamber volume and fipple	Stable A4-ish all-closed tone before finger holes
OCA-P2 4-hole	Prove tuning workflow	Four tuned notes within +/-25 cents after bisque
OCA-P3 12-hole	Full range test	Chromatic pattern playable, no blocked grip
OCA-P4 matched set	Soprano C, Alto C, Bass C	Shared mold logic and glaze family

Open Assumptions

- Pricing in the workbook is an estimate and has not been date-checked.
- Clay shrinkage must be measured for Tony's actual slip, not assumed from supplier average.
- The fipple may need empirical iteration beyond the Helmholtz model.
- Exact ergonomics depend on hand span and final sculptural body shape.

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bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item_id	category	item	qty	spec	make_buy	estimated_cost	source_note	drawing_ref	notes
---	---	---	---	---	---	---	---	---	---
OCA-BOM-001	Master	3D printed body master	1 set	PLA or resin printed top and bottom master	Make	\$2-5	Workbook estimate not date-checked	OCA-DRW-001	Scale master for measured clay shrinkage.
OCA-BOM-002	Mold	#1 pottery plaster	10 lb	USG #1 pottery plaster or equivalent	Buy	\$15-25	Workbook estimate not date-checked	OCA-DRW-002	Enough for several small mold sets depending on cottle size.
OCA-BOM-003	Clay	Cone 6 stoneware slip or clay body	25 lb	Slip casting body or press-moldable clay	Buy	\$20-35	Workbook estimate not date-checked	OCA-DRW-003	Record batch and measured shrinkage.
OCA-BOM-004	Finish	Cone 6 exterior glaze	3-5 pints	Food-safe or durable ceramic glaze	Buy	\$30-60	Workbook estimate not date-checked	OCA-DRW-004	Mask fipple, holes, and interior.
OCA-BOM-005	Tuning	Diamond needle files	1 set	Assorted profiles	Buy	\$10-20	Workbook estimate not date-checked	OCA-DRW-005	Needed after bisque and final fire.
OCA-BOM-006	Tuning	Chromatic tuner or analysis app	1	Cent-accurate tuner	Buy	\$15-30	Workbook estimate not date-checked	OCA-VAL-001	Use same app/settings across prototypes.
OCA-BOM-007	Mold	Mold soap or release	1 bottle	Murphy's oil soap or mold release	Buy	\$5-10	Workbook estimate not date-checked	OCA-DRW-002	For sealed master and plaster parting surfaces.
OCA-BOM-008	Firing	Bisque and glaze firing	per load	Cone 06 bisque and Cone 6 glaze	Buy	\$20-50/load	Workbook estimate not date-checked	OCA-VAL-002	Record firing schedule and kiln location.
OCA-BOM-009	Masking	Wax resist	1 small jar	Brushable wax resist	Buy	TBD	Add supplier/date before purchase	OCA-DRW-004	Protect tuning-critical holes and fipple.
OCA-BOM-010	Measurement	Graduated syringe or burette	1	Water-fill volume measurement	Buy	TBD	Add supplier/date before purchase	OCA-VAL-003	Needed for chamber volume validation.

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sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

| item_id | item | required_spec | search_terms | supplier_candidates | date_checked | unit_price | lead_time | substitution_rule | risk_note |

| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |

| OCA-SRC-001 | #1 pottery plaster | Pottery mold plaster suitable for absorbent ceramic molds | USG #1 pottery plaster 10 lb 25 lb | TBD | | | | Use equivalent pottery plaster if absorption and strength are similar | Wrong plaster can give weak molds or poor casting absorption. |

| OCA-SRC-002 | Cone 6 casting slip | Stoneware slip with known shrinkage and firing schedule | cone 6 stoneware casting slip shrinkage | TBD | | | | Substitute only after measuring shrinkage bars | Unknown shrinkage invalidates CAD scale factor. |

| OCA-SRC-003 | PLA or resin | Print material that can be sanded and sealed | PLA filament smooth master mold making resin print | TBD | | | | Any stable printable material is acceptable if sealed | Layer lines transfer into mold and cast. |

| OCA-SRC-004 | Mold release | Mold soap compatible with plaster | pottery mold soap plaster release | TBD | | | | Commercial mold soap preferred | Release failure can damage master or mold. |

| OCA-SRC-005 | Diamond files | Small round and tapered profiles | diamond needle file set ceramic tuning | TBD | | | | Diamond burrs can substitute for some work | Coarse tools chip fired ceramic. |

| OCA-SRC-006 | Wax resist | Brushable wax for glaze masking | ceramic wax resist glaze masking | TBD | | | | Latex or tape only if kiln-safe workflow is confirmed | Glaze in holes or windway can ruin tuning. |

| OCA-SRC-007 | Measurement tools | Cent-accurate tuner plus volume measurement | chromatic tuner cents water volume graduated syringe | TBD | | | | Phone app acceptable for early tests | Inconsistent measurement hides process drift. |

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cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

| cut_id | part | qty | rough_dimensions_in | final_dimensions_in | material | orientation | operation | tolerance_in | yield_or_offcut | notes |

| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |

| OCA-CUT-001 | Master body upper half (3D-printed) | 1 | 4.50 x 3.50 x 2.50 envelope | master scale = 1/(1-shrinkage); for 12% shrinkage = 1.136x fired body | PLA or sealable resin | X along mouthpiece axis; Z up; mark split-line and chamber datum | Print, sand layer lines, fill voids, seal/sand, sign with build ID | +/-0.010 | Save offcut sprues for shrinkage coupons | Master scale factor must be re-derived from MEASURED clay shrinkage before final master. |

| OCA-CUT-002 | Master body lower half (3D-printed) | 1 | 4.50 x 3.50 x 2.50 envelope | mirror of OCA-CUT-001 | PLA or sealable resin | X along mouthpiece axis; Z up; mark split-line | Print, sand, fill, seal | +/-0.010 | Reuse offcut

sprues for shrinkage coupons | Same scale factor as OCA-CUT-001. |

| OCA-CUT-003 | Wooden cottle boards (mold making) | 4 | 8 x 6 x 0.75 each | 7.50 x 5.50 x 0.75 each | Melamine-faced MDF or sealed plywood | Smooth face inward; mark inner clamping zone | Cut, edge-seal, drill clamping holes | +/-0.030 | One sheet of 24 x 18 x 0.75 yields all 4 | Clean parting wax/release before each pour. |

| OCA-CUT-004 | Plaster mother mold half (poured) | 2 | 7.50 x 5.50 x 2.00 each | 7.50 x 5.50 x 1.50-1.75 each (around master) | USG #1 pottery plaster (or equiv.) | Pour with master keyed to first half; second half over registration | Mix to 73:100 water:plaster (by weight); 2-min slake; pour around sealed master | +/-0.060 | Trim spew/edge after demold; recover small offcuts as test slabs | Min 1.50 in plaster around master where practical for stiffness. |

| OCA-CUT-005 | Greenware vessel (slip-cast) | 1 per pour | ~3.95 x 3.07 x 2.20 (master cavity wet) | ~3.50 x 2.72 x 1.95 fired (12% shrink) | Cone 6 stoneware casting slip | Mouthpiece axis along master split | Slip pour > drain at target wall (~0.16 in / 0.4 cm) > demold leather-hard | N/A (process-controlled) | Reuse demold trim/fettling waste as slip recycle | Wall thickness drives drying time and tap tone — measure on noncritical trim. |

| OCA-CUT-006 | Voicing window cut (leather-hard) | 1 | N/A | 0.354 x 0.197 (window W x H = 0.9 x 0.5 cm per design) | Cone 6 greenware | Cut perpendicular to mouthpiece axis | Score, lift, clean labium edge with rib | +/-0.010 | N/A | Cut undersized; tune by enlarging at bisque tuning step. |

| OCA-CUT-007 | Finger holes (12-hole layout — leather-hard) | 12 | N/A | 0.16-0.31 dia each (start undersized; see design table OCA-DRW-005) | Cone 6 greenware | Per fingering chart row D40:F51 of design table | Brass tube cutter or hole punch; clean burr with damp finger | +/-0.010 (start) | N/A | All 12 holes start at design-table 'minimum' diameter; enlarge at tuning. |

| OCA-CUT-008 | Post-bisque tuning trim | as needed | N/A | Within +/-25 cents of target frequency per validation.csv | Bisqued ceramic | N/A | Diamond needle file or fine diamond burr in steps | +/-0.005 dimensional; +/-25 cents acoustic | Diamond burr offcuts are kiln dust — bag and discard | Document each enlarge step on validation row. |

| OCA-CUT-009 | Wax resist mask (pre-glaze) | 1 application | N/A | ~0.020 in dry film over voicing window + all 12 holes + interior | Brushable wax resist | N/A | Brush-apply; dry per supplier; check all openings clear | Visual | N/A | Mask all acoustic edges to keep glaze out of windway and holes. |

| OCA-CUT-010 | Glaze application (exterior only) | 1 application | N/A | ~0.005-0.010 in per coat; 1-2 coats exterior | Cone 6 satin or matte glaze | Avoid waxed regions | Dip / brush / spray exterior only | +/-0.005 (visual) | N/A | Glaze in voicing window or holes will detune the instrument. |

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drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Ocarina Drawing Brief

Required Views

- Top view with body outline, hole layout, hand assignment, and centerline.
- Side view with mouthpiece axis, windway height, labium angle, body thickness, and split line.
- Section through fipple showing windway, labium, window, and chamber connection.

- Section through body showing chamber volume reference, target wall thickness, and mold split.
- Detail view for hole diameters and post-fire tuning allowance.
- Mold view showing plaster parting line, registration keys, pour/drain plan, and cottle clearance.

Critical Dimensions

Dimension	Source	Tolerance intent
---	---	---
Chamber volume	Workbook input and water-fill measurement	Tuning critical
Voicing window width/height	Workbook input	Playability critical
Windway gap	CAD/test-piece measurement	Playability critical
Labium angle and edge thickness	CAD/test-piece measurement	Playability critical
Wall thickness	Workbook input and cast measurement	Tone/drying critical
Hole diameters	Workbook chart and tuning log	Tuning critical
Master scale factor	Measured clay shrinkage	Process critical

Notes For CAD

- Store chamber volume as a named parameter.
- Store shrinkage factor as a named parameter.
- Separate final fired dimensions from master dimensions.
- Mark fipple geometry as prototype-sensitive; do not freeze it until test pieces speak reliably.
- Include a versioned body ID so castings can be tied back to the CAD master.

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assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Ocarina Assembly Manual

Scope

This manual covers a slip-cast or press-molded ceramic ocarina based on `docs/Ocarina.xlsx`. It is written for prototype builds where the first goal is repeatable tone and validated tuning, not decorative finish.

Tools

- CAD package for master design.
- 3D printer and sanding/sealing supplies.
- Cottle boards, plaster mixing bucket, scale, mold soap, and clamps.
- Stoneware slip or workable clay body.
- Small hole cutters, drill bits, needle tools, loop tools, ribs, sponge, and fettling knife.
- Chromatic tuner, microphone, calipers, graduated syringe or water-fill vessel.
- Diamond needle files or fine diamond burrs.
- Wax resist, glaze, and kiln access.

Process

1. ****Set design inputs****

- Confirm target ocarina size, chamber volume, voicing window, wall thickness, clay body, and shrinkage.
- Create a build ID before CAD begins.

2. ****Model the master****

- Model the body as a controlled-volume vessel.
- Include split-line planning, registration features, and a fipple strategy.
- Scale the master by $\frac{1}{1 - \text{measured_shrinkage}}$.

3. ****Print and finish the master****

- Print the master with enough wall strength to survive mold making.
- Sand layer lines and seal the surface.
- Mark datums: centerline, mouthpiece axis, split plane, and hole reference side.

4. ****Make the mold****

- Apply release to the sealed master and cottle surfaces.
- Pour plaster with at least 1.5 inches around the master where practical.
- Add registration keys and dry the mold fully before casting.

5. ****Cast or press the body****

- For slip casting, pour slip, wait for target wall buildup, drain, and demold at leather-hard stage.
- For press molding, press even clay slabs into both mold halves and join with scored slip.
- Measure sample wall thickness at a noncritical trimmed area.

6. ****Cut fipple and holes****

- Establish the windway first.
- Cut the labium cleanly and test blow before the body fully dries.
- Cut finger holes undersized.

7. ****Dry slowly****

- Dry under plastic until moisture equalizes.
- Watch the mouthpiece and seam areas for cracks.

8. ****Bisque fire****

- Bisque to the clay body's recommended schedule.
- Measure mass, volume, hole diameters, and frequencies.

9. ****Tune after bisque****

- Enlarge holes to raise pitch.
- Refine the fipple only in small steps.
- Record measured frequency and cents error after each change.

10. ****Glaze****

- Wax resist the windway, labium, holes, and any interior opening.
- Keep glaze away from the acoustic edges.

11. ****Final fire and validation****

- Fire to the chosen glaze schedule.
- Re-measure all tuning points.
- Save final dimensions and tuning data back into the validation log.

Failure Modes To Watch

- Weak or no tone: windway too tall, labium too blunt, air jet missing edge.
- Pitch too flat: chamber too large, open area too small, or neck length too long.
- Pitch too sharp: chamber too small, open area too large, or over-fired holes.
- Warped mouthpiece: uneven drying or thin unsupported fipple geometry.
- Glaze-choked note: glaze reduced hole or windway opening.

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validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

build_id	stage	date	clay_body	shrinkage_expected_pct	master_scale_factor	chamber_volume_cm3	wall_thickness_cm	voicing_w_cm	voicing_h_cm	hole_id	target_note	target_freq_hz	measured_freq_hz	cents_error	action	result	notes
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

OCA-P0	voicing_tile		all_closed	A4	440		Practice fipple before full body.				
OCA-P1	greenware		130	0.4	0.9	0.5	all_closed	A4	440		Measure before drying if possible.
OCA-P1	bisque		TBD	TBD	TBD	TBD	all_closed	A4	440		First real model check.
OCA-P1	glaze_fire		TBD	TBD	TBD	TBD	all_closed	A4	440		Record glaze shift.
OCA-P2	bisque		TBD	TBD	TBD	TBD	hole_1	Bb4	466.2		Start undersized and enlarge gradually.
OCA-P2	bisque		TBD	TBD	TBD	TBD	hole_2	B4	493.9		Record hole diameter after tuning.
OCA-P2	bisque		TBD	TBD	TBD	TBD	hole_3	C5	523.3		Check breath pressure sensitivity.
OCA-P3	glaze_fire		TBD	TBD	TBD	TBD	hole_12	A5	880		Full 12-hole validation row.

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supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ — Slip-Cast Ceramic Ocarina (OCA-001 family)

> Use this as a template for outreach to ceramic-supply, plaster, mold-making, and finishing suppliers. Customize the Subject and address fields per recipient.

Subject: RFQ — slip-cast ceramic ocarina prototype materials and consumables

Hello,

I'm a mechanical R&D engineer prototyping a small family of slip-cast ceramic ocarinas (vessel flutes, ~3-4 in long, ~0.16 in / 4 mm wall) and need quotes for the following. The first build is an Alto C 12-hole prototype based on the parametric Helmholtz design table at [tonykoop/ocarina](https://github.com/tonykoop/ocarina).

Items

#	Item	Required spec	Approx. qty (per prototype run)
1	Cone 6 stoneware casting slip	Mature Cone 6 stoneware slip with published shrinkage (X/Y/Z), water absorption % , and firing schedule . White or light buff body preferred for tone consistency over dark iron-bearing slips.	1-5 gal
2	#1 pottery plaster	USG #1 Pottery Plaster (or equivalent) suitable for absorbent slip-cast molds. Need water:plaster ratio , set time , soluble salt content .	10 lb (single mold) — 50 lb (production mold set)
3	Mold release / mold soap	Plaster-compatible parting agent (Murphy's-type oil soap or commercial mold soap).	1 bottle / 16 oz

| 4 | Cone 6 exterior glaze | Food-safe (or durable non-food-safe) satin or matte glaze. Color: TBD per family — likely earth/buff/cream. Glaze must not bridge holes or windway. | 3-5 pints |

| 5 | Brushable wax resist | Standard ceramic wax resist for masking voicing window, finger holes, and chamber interior pre-glaze. | 1 small jar / 8 oz |

| 6 | Diamond needle file set (post-bisque tuning) | Assorted profiles (round, half-round, taper) ~3-6 mm; resin-bonded acceptable. | 1 set |

| 7 | Hole cutters / brass tube cutters | Round cutters from ~0.156 in (4 mm) to ~0.313 in (8 mm) diameter for greenware finger-hole cutting. | 1 set |

| 8 | Bisque + glaze firing service (if local) | Cone 06 bisque + Cone 6 glaze fire; small cubic-foot kiln load. | 2 fires per prototype |

| 9 | (Optional) Master printing/outsource | If you do high-resolution PLA or resin prints up to ~5 in long with **0.005 in dimensional accuracy**, please quote. We currently print in-house but want a backup. | 1-5 master pairs |

What we need in your quote

- **Unit price** and any volume breaks
- **Minimum order quantity**
- **Current stock status** and **lead time**
- **Shipping estimate** to 94566 (Pleasanton, CA) — or local pickup option
- **Material safety data sheet (MSDS)** and **technical data sheet** for items 1, 2, 4, 5
- **Recommended substitutions** for thin-walled (~4 mm) musical instrument bodies
- For item 1: most recent **shrinkage test data** for your batch (we will re-measure on our side, but a baseline matters)

Acceptance criteria

The prototypes are measured for X/Y/Z shrinkage, bore/cavity volume after firing, wall-thickness consistency, and acoustic tuning (chromatic tuner, ± 25 cents target post-bisque). Slip with unknown or wildly variable shrinkage invalidates the master scale factor (currently $1/(1-0.12) = 1.136$), so reproducibility data is more valuable than headline price.

The repo is a private R&D portfolio repo; we do not need any IP-restricted documents. Public-domain MSDS and supplier spec sheets are sufficient.

Thank you,

Tony Koop

Mechanical R&D Engineer

tonykoop@gmail.com

github.com/tonykoop/ocarina (private; access on request)

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visual-bom-brief.md

Art direction for an image-forward visual BOM.

Ocarina Visual BOM Brief

Goal

Create a one-page visual BOM that helps a builder understand the ceramic ocarina workflow from digital master to tuned fired instrument.

Layout

- Header: "Slip-Cast Ceramic Ocarina - Alto C Prototype".
- Hero image: finished ocarina or clean CAD render.
- Process strip: CAD master, 3D print, plaster mold, greenware, bisque, tuning, glaze fire.
- BOM table: item number, part/material, quantity, spec, estimated cost, make/buy, image.
- Callout inset: fipple section with windway, labium, and voicing window labels.
- Validation inset: chamber volume, all-closed note, highest tested note, cents error.

Image Requirements

Use real shop photos when available. Generated or rendered placeholders should be labeled as placeholders until replaced with:

- Master print photo.
- Plaster mold photo.
- Greenware body photo.
- Fipple close-up.
- Final fired instrument.
- Tuning tool/photo of measurement setup.

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wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

(* Ocarina Helmholtz notebook starter *)

```

ClearAll["Global`*"];

c = 34300; (* cm/s at about 20 C *)
targetA4 = 440;
volumeCm3 = 130;
wallCm = 0.4;
windowWcm = 0.9;
windowHcm = 0.5;

voicingAreaCm2[w_, h_] := w*h;
endCorrectionCm[area_] := 0.6*Sqrt[area/Pi];
helmholtzHz[area_, volume_, neckLength_] :=
(c/(2*Pi))*Sqrt[area/(volume*neckLength)];
centsError[measured_, target_] := 1200*Log[2, measured/target];

baseArea = voicingAreaCm2[windowWcm, windowHcm];
baseNeck = wallCm + endCorrectionCm[baseArea];
baseFreq = helmholtzHz[baseArea, volumeCm3, baseNeck];

holeDiameterMmToAreaCm2[d_] := Pi*(d/20)^2;
targetFreq[midi_] := 440*2^((midi - 69)/12);
areaForFrequency[f_, volume_, neckLength_] :=
volume*neckLength*(2*Pi*f/c)^2;

(* Candidate cumulative area table for A4 to A5. *)
notes = {
{"A4", 69}, {"Bb4", 70}, {"B4", 71}, {"C5", 72},
{"C#5", 73}, {"D5", 74}, {"D#5", 75}, {"E5", 76},
{"F5", 77}, {"F#5", 78}, {"G5", 79}, {"G#5", 80}, {"A5", 81}
};

areaTable = Table[
With[{freq = targetFreq[midi]},
{name, freq, areaForFrequency[freq, volumeCm3, baseNeck]}
],
{ {name, midi}, notes}
];

areaTable // TableForm

```

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README.md

Project artifact.

Ocarina — Slip-Cast Ceramic Vessel-Flute Family

> *Engineering documentation for a 3D-printed-master, plaster-mold, cone-6-stoneware slip-cast ocarina family — from Helmholtz physics through the parametric design table to a manufacturable build packet.*

![[Hero — AI-generated concept render of a slip-cast Alto C 12-hole ocarina (placeholder until first prototype is photographed)]](images/hero-concept.

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capstone-deck.md

Project artifact.

Slip-Cast Ceramic Ocarina (OCA-001) — v3 Smoke Test

- Musical instrument documentation capstone
- Build packet: ocarina
- Generated: 2026-05-02

Project Intent

- Build a slip-cast ceramic Alto C 12-hole ocarina using a 3D-printed master and plaster mold workflow. The current table is focused on the standard "sweet potato" vessel-flute form, but the same model can be used as the baseline for sculptural vessel flutes.

Speaker notes: Read design.md before committing to dimensions or sourcing decisions.

Physics Model

- Ocarinas behave primarily as Helmholtz resonators:

$$f = c / (2 * \pi) * \sqrt{A_{\text{open}} / (V_{\text{chamber}} * L_{\text{eff}})}$$

Speaker notes: Governing equation extracted verbatim from design.md. Apply empirical corrections (NAF K2, scale offsets) only where the model permits — see references/acoustic-models.md.

How To Use This Packet

- Start with design.md for intent and assumptions.
- Use bom.csv, sourcing.csv, and cut-list.csv before buying or cutting.
- Use drawing-brief.md and CAD/CNC folders before machining.
- Print the packet for shopping, shop work, and validation.

File Map

- design.md: Project intent, catalog metadata, assumptions, and validation plan.
- bom.csv: Starter bill of materials with part categories, quantities, drawing refs, and notes.
- sourcing.csv: Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
- cut-list.csv: Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
- drawing-brief.md: Manufacturing drawing and technical product sketch brief.
- assembly-manual.md: Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
- validation.csv: Target/measured values, tolerance, environment, result, and tuning/build action log.
- supplier-rfq.md: Supplier email/request-for-quote starter.

Build Workflow

- Design and assumptions
- Source materials and hardware
- Prepare stock, fixtures, and CNC/laser/lathe setup
- Assemble, tune, finish, and validate

Sourcing And BOM

- BOM gives part categories and drawing references.
- Sourcing tracks search terms, supplier candidates, price/date, lead time, substitutions.
- Visual BOM brief turns the parts list into a presentation-ready image board.

Shop Packet

- Cut list for lumber/sheet/blank planning.

- Assembly manual for away-from-keyboard work.
- Validation sheet for measured dimensions, tuning, pass/fail checks.

Drawings, CAD, CNC

- drawing-brief.md defines required views, dimensions, datums, sketch intent.
- cad/ holds models and design tables.
- cnc/ holds CAM, toolpaths, setup sheets, dry-run notes.
- drawings/ holds PDFs, SVGs, DXFs, drawing exports.

Images And Screenshots

- images/exploded-view-concept.png
- images/hero-concept.png
- images/mold-cross-section-concept.png

Validation Plan

- A4 = 440 Hz reference check.
- Tuning targets logged in validation.csv.
- Critical dimensions verified against design sheet and CAD.
- Photos and revision notes after each major step.

Open Risks / Decisions

- TBDs in design sheet and BOM.
- Supplier price/availability not yet verified.
- Generated images marked as concept placeholders.

- Empirical corrections await measured prototype data.

Next Actions

- Replace TBDs with measured/source-backed values.
- Verify live supplier price and availability before buying.
- Export final drawings and visual BOM images.
- Regenerate this deck and print packet after final edits.

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print-packet.md

Project artifact.

Slip-Cast Ceramic Ocarina (OCA-001) — Build Packet Print Packet

Generated: 2026-05-02

Packet folder: `sessions/charming-bold-fermat/mnt/GitHub/ocarina`

File Map

| File | Purpose |

| --- | --- |

| `design.md` | Project intent, catalog metadata, assumptions, and validation plan. |

| `bom.csv` | Starter bill of materials with part categories, quantities, drawing refs, and notes. |

| `sourcing.csv` | Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks. |

| `cut-list.csv` | Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts. |

| `drawing-brief.md` | Manufacturing drawing and technical product sketch brief. |

| `assembly-manual.md` | Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes. |

| `validation.csv` | Target/measured values, tolerance, environment, result, and tuning/build action log. |

| `supplier-rfq.md` | Supplier email/request-for-quote starter. |

| `visual-bom-brief.md` | Art direction for an image-forward visual BOM. |

| `wolfram-starter.wl` | Wolfram starter for physics, optimization, visualization, and validation. |

| `README.md` | Project artifact. |

| `capstone-deck.md` | Project artifact. |
| `print-packet.md` | Project artifact. |

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design.md

Project intent, catalog metadata, assumptions, and validation plan.

Ocarina Build Packet

Source

- Design table: `docs/Ocarina.xlsx`
- Sheet: `Ocarina`
- Inspected range: `A1:H141`
- Workbook content observed: design inputs, size library, Helmholtz calculator, 12-hole fingering chart, ceramic workflow, BOM, design notes, and Wolfram notebook notes.

Design Intent

Build a slip-cast ceramic Alto C 12-hole ocarina using a 3D-printed master and plaster mold workflow. The current table is focused on the standard "sweet potato" vessel-flute form, but the same model can be used as the baseline for sculptural vessel flutes.

Governing Model

Ocarinas behave primarily as Helmholtz resonators:

$$f = c/(2\pi) * \sqrt{A_{\text{open}}/(V_{\text{chamber}} * L_{\text{eff}})}$$

Unlike a transverse flute or Native American style flute, pitch is driven by chamber volume and total open-hole area, not tone-hole distance along a bore. Hole position is still important for ergonomics, grip, and fingering logic.

Current Workbook Inputs

Field	Current value
Ocarina type	Alto C, 12-hole
Chamber volume	130 cm³
Wall thickness	0.4 cm

Voicing window width	0.9 cm
Voicing window height	0.5 cm
Speed of sound	34300 cm/s
Clay body	Cone 6 stoneware
Shrinkage	12 percent
Target low note	A4, 440 Hz
Target high note	F6, 1397 Hz

Critical Design Features

- Chamber volume is the master tuning input. Measure it by water fill after CAD, master print, greenware, bisque, and glaze fire.
- Voicing window area sets the all-closed low-note behavior along with volume and effective neck length.
- Fipple geometry is the playability bottleneck: windway gap, labium angle, channel smoothness, and edge sharpness should be treated as first-class dimensions.
- Finger holes should start undersized. Enlarge to sharpen; flattening after over-enlarging is possible but slower and less clean.
- Exterior-only glazing is preferred for tuning stability and acoustic response. Mask the fipple, holes, and chamber interior.

Workbook Improvement Notes

During inspection, rows 39-42 of the 12-hole fingering chart contain formulas for hole area, cumulative open area, and calculated frequency. Rows 43-51 currently show placeholder `=` cells instead of filled formulas. Recommended next workbook cleanup:

1. Copy the area formula pattern from `D40:D42` through `D51`.
2. Copy the cumulative area pattern from `E40:E42` through `E51`.
3. Copy the frequency formula pattern from `F40:F42` through `F51`.
4. Add measured frequency, cents error, and tuning action columns.
5. Add a note that hole combinations can be cross-fingered if total open area is equivalent.
6. Verify the size-library `Length (in)` column. Inspection showed serial-like values such as `46180`, so those cells may have been formatted or entered incorrectly.

Mold Strategy

Use a 2-piece plaster mold for the main body. The first prototype can use a press-molded body to speed iteration, but production should move toward slip casting once the fipple and chamber volume are stable.

Recommended mold/CAD features:

- Split line around the widest body perimeter.
- Positive registration keys that do not trap clay.
- Pour/drain path or assembly opening appropriate to the chosen casting workflow.
- Removable or cleanable fipple detail. If the fipple is too delicate for the mold, cast a blank and hand-cut the voicing at leather-hard stage.
- Master scale factor: $\frac{1}{(1 - \text{shrinkage})}$; for 12 percent shrinkage this is about 1.136.

Prototype Ladder

Prototype	Goal	Success criteria
---	---	---
OCA-P0 voicing tile	Practice windway/labium only	Clear tone on a disposable test cavity
OCA-P1 closed vessel	Verify chamber volume and fipple	Stable A4-ish all-closed tone before finger holes
OCA-P2 4-hole	Prove tuning workflow	Four tuned notes within +/-25 cents after bisque
OCA-P3 12-hole	Full range test	Chromatic pattern playable, no blocked grip
OCA-P4 matched set	Soprano C, Alto C, Bass C	Shared mold logic and glaze family

Open Assumptions

- Pricing in the workbook is an estimate and has not been date-checked.
- Clay shrinkage must be measured for Tony's actual slip, not assumed from supplier average.
- The fipple may need empirical iteration beyond the Helmholtz model.
- Exact ergonomics depend on hand span and final sculptural body shape.

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bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item_id	category	item	qty	spec	make_buy	estimated_cost	source_note	drawing_ref	notes
---	---	---	---	---	---	---	---	---	---
OCA-BOM-001	Master	3D printed body master	1 set	PLA or resin printed top and bottom master	Make	\$2-5	Workbook estimate not date-checked	OCA-DRW-001	Scale master for measured clay shrinkage.
OCA-BOM-002	Mold	#1 pottery plaster	10 lb	USG #1 pottery plaster or equivalent	Buy	\$15-25	Workbook estimate not date-checked	OCA-DRW-002	Enough for several small mold sets depending on cottle size.
OCA-BOM-003	Clay	Cone 6 stoneware slip or clay body	25 lb	Slip casting body or press-moldable clay	Buy	\$20-35	Workbook estimate not date-checked	OCA-DRW-003	Record batch and measured shrinkage.
OCA-BOM-004	Finish	Cone 6 exterior glaze	3-5 pints	Food-safe or durable ceramic glaze	Buy	\$30-60	Workbook estimate not date-checked	OCA-DRW-004	Mask fipple, holes, and interior.

| OCA-BOM-005 | Tuning | Diamond needle files | 1 set | Assorted profiles | Buy | \$10-20 | Workbook estimate not date-checked | OCA-DRW-005 | Needed after bisque and final fire. |

| OCA-BOM-006 | Tuning | Chromatic tuner or analysis app | 1 | Cent-accurate tuner | Buy | \$15-30 | Workbook estimate not date-checked | OCA-VAL-001 | Use same app/settings across prototypes. |

| OCA-BOM-007 | Mold | Mold soap or release | 1 bottle | Murphy's oil soap or mold release | Buy | \$5-10 | Workbook estimate not date-checked | OCA-DRW-002 | For sealed master and plaster parting surfaces. |

| OCA-BOM-008 | Firing | Bisque and glaze firing | per load | Cone 06 bisque and Cone 6 glaze | Buy | \$20-50/load | Workbook estimate not date-checked | OCA-VAL-002 | Record firing schedule and kiln location. |

| OCA-BOM-009 | Masking | Wax resist | 1 small jar | Brushable wax resist | Buy | TBD | Add supplier/date before purchase | OCA-DRW-004 | Protect tuning-critical holes and fipple. |

| OCA-BOM-010 | Measurement | Graduated syringe or burette | 1 | Water-fill volume measurement | Buy | TBD | Add supplier/date before purchase | OCA-VAL-003 | Needed for chamber volume validation. |

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sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

item_id	item	required_spec	search_terms	supplier_candidates	date_checked	unit_price	lead_time	substitution_rule	risk_note
---	---	---	---	---	---	---	---	---	---
OCA-SRC-001	#1 pottery plaster	Pottery mold plaster suitable for absorbent ceramic molds	USG #1 pottery plaster 10 lb 25 lb	TBD		Use equivalent pottery plaster if absorption and strength are similar	Wrong plaster can give weak molds or poor casting absorption.		
OCA-SRC-002	Cone 6 casting slip	Stoneware slip with known shrinkage and firing schedule	cone 6 stoneware casting slip shrinkage	TBD		Substitute only after measuring shrinkage bars	Unknown shrinkage invalidates CAD scale factor.		
OCA-SRC-003	PLA or resin	Print material that can be sanded and sealed	PLA filament smooth master mold making resin print	TBD		Any stable printable material is acceptable if sealed	Layer lines transfer into mold and cast.		
OCA-SRC-004	Mold release	Mold soap compatible with plaster	pottery mold soap plaster release	TBD		Commercial mold soap preferred	Release failure can damage master or mold.		
OCA-SRC-005	Diamond files	Small round and tapered profiles	diamond needle file set ceramic tuning	TBD		Diamond burrs can substitute for some work	Coarse tools chip fired ceramic.		
OCA-SRC-006	Wax resist	Brushable wax for glaze masking	ceramic wax resist glaze masking	TBD		Latex or tape only if kiln-safe workflow is confirmed	Glaze in holes or windway can ruin tuning.		
OCA-SRC-007	Measurement tools	Cent-accurate tuner plus volume measurement	chromatic tuner cents water volume graduated syringe	TBD		Phone app acceptable for early tests	Inconsistent measurement hides process drift.		

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cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

| cut_id | part | qty | rough_dimensions_in | final_dimensions_in | material | orientation | operation | tolerance_in | yield_or_offcut | notes |

| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |

| OCA-CUT-001 | Master body upper half (3D-printed) | 1 | 4.50 x 3.50 x 2.50 envelope | master scale = 1/(1-shrinkage); for 12% shrinkage = 1.136x fired body | PLA or sealable resin | X along mouthpiece axis; Z up; mark split-line and chamber datum | Print, sand layer lines, fill voids, seal/sand, sign with build ID | +/-0.010 | Save offcut sprues for shrinkage coupons | Master scale factor must be re-derived from MEASURED clay shrinkage before final master. |

| OCA-CUT-002 | Master body lower half (3D-printed) | 1 | 4.50 x 3.50 x 2.50 envelope | mirror of OCA-CUT-001 | PLA or sealable resin | X along mouthpiece axis; Z up; mark split-line | Print, sand, fill, seal | +/-0.010 | Reuse offcut sprues for shrinkage coupons | Same scale factor as OCA-CUT-001. |

| OCA-CUT-003 | Wooden cottle boards (mold making) | 4 | 8 x 6 x 0.75 each | 7.50 x 5.50 x 0.75 each | Melamine-faced MDF or sealed plywood | Smooth face inward; mark inner clamping zone | Cut, edge-seal, drill clamping holes | +/-0.030 | One sheet of 24 x 18 x 0.75 yields all 4 | Clean parting wax/release before each pour. |

| OCA-CUT-004 | Plaster mother mold half (poured) | 2 | 7.50 x 5.50 x 2.00 each | 7.50 x 5.50 x 1.50-1.75 each (around master) | USG #1 pottery plaster (or equiv.) | Pour with master keyed to first half; second half over registration | Mix to 73:100 water:plaster (by weight); 2-min slake; pour around sealed master | +/-0.060 | Trim spew/edge after demold; recover small offcuts as test slabs | Min 1.50 in plaster around master where practical for stiffness. |

| OCA-CUT-005 | Greenware vessel (slip-cast) | 1 per pour | ~3.95 x 3.07 x 2.20 (master cavity wet) | ~3.50 x 2.72 x 1.95 fired (12% shrink) | Cone 6 stoneware casting slip | Mouthpiece axis along master split | Slip pour > drain at target wall (~0.16 in / 0.4 cm) > demold leather-hard | N/A (process-controlled) | Reuse demold trim/fettling waste as slip recycle | Wall thickness drives drying time and tap tone — measure on noncritical trim. |

| OCA-CUT-006 | Voicing window cut (leather-hard) | 1 | N/A | 0.354 x 0.197 (window W x H = 0.9 x 0.5 cm per design) | Cone 6 greenware | Cut perpendicular to mouthpiece axis | Score, lift, clean labium edge with rib | +/-0.010 | N/A | Cut undersized; tune by enlarging at bisque tuning step. |

| OCA-CUT-007 | Finger holes (12-hole layout — leather-hard) | 12 | N/A | 0.16-0.31 dia each (start undersized; see design table OCA-DRW-005) | Cone 6 greenware | Per fingering chart row D40:F51 of design table | Brass tube cutter or hole punch; clean burr with damp finger | +/-0.010 (start) | N/A | All 12 holes start at design-table 'minimum' diameter; enlarge at tuning. |

| OCA-CUT-008 | Post-bisque tuning trim | as needed | N/A | Within +/-25 cents of target frequency per validation.csv | Bisqued ceramic | N/A | Diamond needle file or fine diamond burr in steps | +/-0.005 dimensional; +/-25 cents acoustic | Diamond burr offcuts are kiln dust — bag and discard | Document each enlarge step on validation row. |

| OCA-CUT-009 | Wax resist mask (pre-glaze) | 1 application | N/A | ~0.020 in dry film over voicing window + all 12 holes + interior | Brushable wax resist | N/A | Brush-apply; dry per supplier; check all openings clear | Visual | N/A | Mask all acoustic edges to keep glaze out of windway and holes. |

| OCA-CUT-010 | Glaze application (exterior only) | 1 application | N/A | ~0.005-0.010 in per coat; 1-2 coats exterior | Cone 6 satin or matte glaze | Avoid waxed regions | Dip / brush / spray exterior only | +/-0.005 (visual) | N/A | Glaze in voicing window or holes will detune the instrument. |

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drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Ocarina Drawing Brief

Required Views

- Top view with body outline, hole layout, hand assignment, and centerline.
- Side view with mouthpiece axis, windway height, labium angle, body thickness, and split line.
- Section through fipple showing windway, labium, window, and chamber connection.
- Section through body showing chamber volume reference, target wall thickness, and mold split.
- Detail view for hole diameters and post-fire tuning allowance.
- Mold view showing plaster parting line, registration keys, pour/drain plan, and cottle clearance.

Critical Dimensions

Dimension Source Tolerance intent
--- --- ---
Chamber volume Workbook input and water-fill measurement Tuning critical
Voicing window width/height Workbook input Playability critical
Windway gap CAD/test-piece measurement Playability critical
Labium angle and edge thickness CAD/test-piece measurement Playability critical
Wall thickness Workbook input and cast measurement Tone/drying critical
Hole diameters Workbook chart and tuning log Tuning critical
Master scale factor Measured clay shrinkage Process critical

Notes For CAD

- Store chamber volume as a named parameter.
- Store shrinkage factor as a named parameter.
- Separate final fired dimensions from master dimensions.
- Mark fipple geometry as prototype-sensitive; do not freeze it until test pieces speak reliably.
- Include a versioned body ID so castings can be tied back to the CAD master.

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assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Ocarina Assembly Manual

Scope

This manual covers a slip-cast or press-molded ceramic ocarina based on `docs/Ocarina.xlsx`. It is written for prototype builds where the first goal is repeatable tone and validated tuning, not decorative finish.

Tools

- CAD package for master design.
- 3D printer and sanding/sealing supplies.
- Cottle boards, plaster mixing bucket, scale, mold soap, and clamps.
- Stoneware slip or workable clay body.
- Small hole cutters, drill bits, needle tools, loop tools, ribs, sponge, and fettling knife.
- Chromatic tuner, microphone, calipers, graduated syringe or water-fill vessel.
- Diamond needle files or fine diamond burrs.
- Wax resist, glaze, and kiln access.

Process

1. ****Set design inputs****
 - Confirm target ocarina size, chamber volume, voicing window, wall thickness, clay body, and shrinkage.
 - Create a build ID before CAD begins.
2. ****Model the master****
 - Model the body as a controlled-volume vessel.
 - Include split-line planning, registration features, and a fipple strategy.
 - Scale the master by $\frac{1}{1 - \text{measured_shrinkage}}$.
3. ****Print and finish the master****
 - Print the master with enough wall strength to survive mold making.
 - Sand layer lines and seal the surface.
 - Mark datums: centerline, mouthpiece axis, split plane, and hole reference side.
4. ****Make the mold****
 - Apply release to the sealed master and cottle surfaces.

- Pour plaster with at least 1.5 inches around the master where practical.
 - Add registration keys and dry the mold fully before casting.
5. ****Cast or press the body****
- For slip casting, pour slip, wait for target wall buildup, drain, and demold at leather-hard stage.
 - For press molding, press even clay slabs into both mold halves and join with scored slip.
 - Measure sample wall thickness at a noncritical trimmed area.
6. ****Cut fipple and holes****
- Establish the windway first.
 - Cut the labium cleanly and test blow before the body fully dries.
 - Cut finger holes undersized.
7. ****Dry slowly****
- Dry under plastic until moisture equalizes.
 - Watch the mouthpiece and seam areas for cracks.
8. ****Bisque fire****
- Bisque to the clay body's recommended schedule.
 - Measure mass, volume, hole diameters, and frequencies.
9. ****Tune after bisque****
- Enlarge holes to raise pitch.
 - Refine the fipple only in small steps.
 - Record measured frequency and cents error after each change.
10. ****Glaze****
- Wax resist the windway, labium, holes, and any interior opening.
 - Keep glaze away from the acoustic edges.
11. ****Final fire and validation****
- Fire to the chosen glaze schedule.
 - Re-measure all tuning points.
 - Save final dimensions and tuning data back into the validation log.

Failure Modes To Watch

- Weak or no tone: windway too tall, labium too blunt, air jet missing edge.
- Pitch too flat: chamber too large, open area too small, or neck length too long.
- Pitch too sharp: chamber too small, open area too large, or over-filed holes.

- Warped mouthpiece: uneven drying or thin unsupported fipple geometry.
- Glaze-choked note: glaze reduced hole or windway opening.

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validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

```
| build_id | stage | date | clay_body | shrinkage_expected_pct | master_scale_factor | chamber_volume_cm3 |
wall_thickness_cm | voicing_w_cm | voicing_h_cm | hole_id | target_note | target_freq_hz | measured_freq_hz |
cents_error | action | result | notes |

| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| OCA-P0 | voicing_tile | | | | | | | all_closed | A4 | 440 | | | | Practice fipple before full body. |
| OCA-P1 | greenware | | | | 130 | 0.4 | 0.9 | 0.5 | all_closed | A4 | 440 | | | | Measure before drying if possible. |
| OCA-P1 | bisque | | | | TBD | TBD | TBD | TBD | all_closed | A4 | 440 | | | | First real model check. |
| OCA-P1 | glaze_fire | | | | TBD | TBD | TBD | TBD | all_closed | A4 | 440 | | | | Record glaze shift. |
| OCA-P2 | bisque | | | | TBD | TBD | TBD | TBD | hole_1 | Bb4 | 466.2 | | | | Start undersized and enlarge gradually.
|
| OCA-P2 | bisque | | | | TBD | TBD | TBD | TBD | hole_2 | B4 | 493.9 | | | | Record hole diameter after tuning. |
| OCA-P2 | bisque | | | | TBD | TBD | TBD | TBD | hole_3 | C5 | 523.3 | | | | Check breath pressure sensitivity. |
| OCA-P3 | glaze_fire | | | | TBD | TBD | TBD | TBD | hole_12 | A5 | 880 | | | | Full 12-hole validation row. |
```

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supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ — Slip-Cast Ceramic Ocarina (OCA-001 family)

> Use this as a template for outreach to ceramic-supply, plaster, mold-making, and finishing suppliers. Customize the Subject and address fields per recipient.

****Subject:**** RFQ — slip-cast ceramic ocarina prototype materials and consumables

Hello,

I'm a mechanical R&D engineer prototyping a small family of slip-cast ceramic ocarinas (vessel flutes, ~3-4 in long, ~0.16 in / 4 mm wall) and need quotes for the following. The first build is an Alto C 12-hole prototype based on the parametric Helmholtz design table at [tonykoop/ocarina](https://github.com/tonykoop/ocarina).

Items

| # | Item | Required spec | Approx. qty (per prototype run) |

| ---|---|---|---|

| 1 | Cone 6 stoneware casting slip | Mature Cone 6 stoneware slip with **published shrinkage** (X/Y/Z), **water absorption %**, and **firing schedule**. White or light buff body preferred for tone consistency over dark iron-bearing slips. | 1-5 gal |

| 2 | #1 pottery plaster | USG #1 Pottery Plaster (or equivalent) suitable for absorbent slip-cast molds. Need **water:plaster ratio**, **set time**, **soluble salt content**. | 10 lb (single mold) — 50 lb (production mold set) |

| 3 | Mold release / mold soap | Plaster-compatible parting agent (Murphy's-type oil soap or commercial mold soap). | 1 bottle / 16 oz |

| 4 | Cone 6 exterior glaze | Food-safe (or durable non-food-safe) satin or matte glaze. Color: TBD per family — likely earth/buff/cream. Glaze must not bridge holes or windway. | 3-5 pints |

| 5 | Brushable wax resist | Standard ceramic wax resist for masking voicing window, finger holes, and chamber interior pre-glaze. | 1 small jar / 8 oz |

| 6 | Diamond needle file set (post-bisque tuning) | Assorted profiles (round, half-round, taper) ~3-6 mm; resin-bonded acceptable. | 1 set |

| 7 | Hole cutters / brass tube cutters | Round cutters from ~0.156 in (4 mm) to ~0.313 in (8 mm) diameter for greenware finger-hole cutting. | 1 set |

| 8 | Bisque + glaze firing service (if local) | Cone 06 bisque + Cone 6 glaze fire; small cubic-foot kiln load. | 2 fires per prototype |

| 9 | (Optional) Master printing/outsource | If you do high-resolution PLA or resin prints up to ~5 in long with **0.005 in dimensional accuracy**, please quote. We currently print in-house but want a backup. | 1-5 master pairs |

What we need in your quote

- **Unit price** and any volume breaks
- **Minimum order quantity**
- **Current stock status** and **lead time**
- **Shipping estimate** to 94566 (Pleasanton, CA) — or local pickup option
- **Material safety data sheet (MSDS)** and **technical data sheet** for items 1, 2, 4, 5
- **Recommended substitutions** for thin-walled (~4 mm) musical instrument bodies
- For item 1: most recent **shrinkage test data** for your batch (we will re-measure on our side, but a baseline matters)

Acceptance criteria

The prototypes are measured for X/Y/Z shrinkage, bore/cavity volume after firing, wall-thickness consistency, and acoustic tuning (chromatic tuner, ± 25 cents target post-bisque). Slip with unknown or wildly variable shrinkage invalidates the master scale factor (currently $\frac{1}{1-0.12} = 1.136$), so reproducibility data is more valuable than headline price.

The repo is a private R&D portfolio repo; we do not need any IP-restricted documents. Public-domain MSDS and supplier spec sheets are sufficient.

Thank you,

Tony Koop

Mechanical R&D Engineer

tonykoop@gmail.com

github.com/tonykoop/ocarina (private; access on request)

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visual-bom-brief.md

Art direction for an image-forward visual BOM.

Ocarina Visual BOM Brief

Goal

Create a one-page visual BOM that helps a builder understand the ceramic ocarina workflow from digital master to tuned fired instrument.

Layout

- Header: "Slip-Cast Ceramic Ocarina - Alto C Prototype".
- Hero image: finished ocarina or clean CAD render.
- Process strip: CAD master, 3D print, plaster mold, greenware, bisque, tuning, glaze fire.
- BOM table: item number, part/material, quantity, spec, estimated cost, make/buy, image.
- Callout inset: fipple section with windway, labium, and voicing window labels.
- Validation inset: chamber volume, all-closed note, highest tested note, cents error.

Image Requirements

Use real shop photos when available. Generated or rendered placeholders should be labeled as placeholders until replaced with:

- Master print photo.
- Plaster mold photo.
- Greenware body photo.
- Fipple close-up.

- Final fired instrument.
- Tuning tool/photo of measurement setup.

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wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

(* Ocarina Helmholtz notebook starter *)

```
ClearAll["Global`*"];

c = 34300; (* cm/s at about 20 C *)
targetA4 = 440;
volumeCm3 = 130;
wallCm = 0.4;
windowWcm = 0.9;
windowHcm = 0.5;

voicingAreaCm2[w_, h_] := w*h;
endCorrectionCm[area_] := 0.6*Sqrt[area/Pi];
helmholtzHz[area_, volume_, neckLength_] :=
(c/(2*Pi))*Sqrt[area/(volume*neckLength)];
centsError[measured_, target_] := 1200*Log[2, measured/target];

baseArea = voicingAreaCm2[windowWcm, windowHcm];
baseNeck = wallCm + endCorrectionCm[baseArea];
baseFreq = helmholtzHz[baseArea, volumeCm3, baseNeck];

holeDiameterMmToAreaCm2[d_] := Pi*(d/20)^2;
targetFreq[midi_] := 440*2^((midi - 69)/12);
areaForFrequency[f_, volume_, neckLength_] :=
volume*neckLength*(2*Pi*f/c)^2;

(* Candidate cumulative area table for A4 to A5. *)
notes = {
{"A4", 69}, {"Bb4", 70}, {"B4", 71}, {"C5", 72},
{"C#5", 73}, {"D5", 74}, {"D#5", 75}, {"E5", 76},
{"F5", 77}, {"F#5", 78}, {"G5", 79}, {"G#5", 80}, {"A5", 81}
};
```

```
areaTable = Table[
  With[{freq = targetFreq[midi]},
    {name, freq, areaForFrequency[freq, volumeCm3, baseNeck]}
  ],
  { {name, midi}, notes}
];
```

```
areaTable // TableForm
```

```
<div class="page-break"></div>
```

README.md

Project artifact.

Ocarina — Slip-Cast Ceramic Vessel-Flute Family

> *Engineering documentation for a 3D-printed-master, plaster-mold, cone-6-stoneware slip-cast ocarina family — from Helmholtz physics through the parametric design table to a manufacturable build packet.*

![[Hero — AI-generated concept render of a slip-cast Alto C 12-hole ocarina (placeholder until first prototype is photographed)]](images/hero-concept.

```
<div class="page-break"></div>
```

capstone-deck.md

Project artifact.

Slip-Cast Ceramic Ocarina (OCA-001) — Build Packet

- Musical instrument documentation capstone
- Build packet: ocarina
- Generated: 2026-05-02

Project Intent

- See design.md for intent.

Speaker notes: Read design.md before committing to dimensions or sourcing decisions.

How To Use This Packet

- Start with design.md for intent and assumptions.
- Use bom.csv, sourcing.csv, and cut-list.csv before buying or cutting.
- Use drawing-brief.md and CAD/CNC folders before machining.
- Print the packet for shopping, shop work, and validation.

File Map

- design.md: Project intent, catalog metadata, assumptions, and validation plan.
- bom.csv: Starter bill of materials with part categories, quantities, drawing refs, and notes.
- sourcing.csv: Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
- cut-list.csv: Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
- drawing-brief.md: Manufacturing drawing and technical product sketch brief.
- assembly-manual.md: Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
- validation.csv: Target/measured values, tolerance, environment, result, and tuning/build action log.
- supplier-rfq.md: Supplier email/request-for-quote starter.

Build Workflow

- Design and assumptions
- Source materials and hardware
- Prepare stock, fixtures, and CNC/laser/lathe setup
- Assemble, tune, finish, and validate

Sourcing And BOM

- BOM gives part categories and drawing references.
- Sourcing tracks search terms, supplier candidates, price/date, lead time, substitutions.
- Visual BOM brief turns the parts list into a presentation-ready image board.

Shop Packet

- Cut list for lumber/sheet/blank planning.
- Assembly manual for away-from-keyboard work.

- Validation sheet for measured dimensions, tuning, pass/fail checks.

Drawings, CAD, CNC

- drawing-brief.md defines required views, dimensions, datums, sketch intent.
- cad/ holds models and design tables.
- cnc/ holds CAM, toolpaths, setup sheets, dry-run notes.
- drawings/ holds PDFs, SVGs, DXFs, drawing exports.

Images And Screenshots

- images/exploded-view-concept.png
- images/hero-concept.png
- images/mold-cross-section-concept.png

Validation Plan

- A4 = 440 Hz reference check.
- Tuning targets logged in validation.csv.
- Critical dimensions verified against design sheet and CAD.
- Photos and revision notes after each major step.

Open Risks / Decisions

- TBDs in design sheet and BOM.
- Supplier price/availability not yet verified.
- Generated images marked as concept placeholders.
- Empirical corrections await measured prototype data.

Next Actions

- Replace TBDs with measured/source-backed values.
- Verify live supplier price and availability before buying.
- Export final drawings and visual BOM images.

- Regenerate this deck and print packet after final edits.

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print-packet.md

Project artifact.

Slip-Cast Ceramic Ocarina (OCA-001) — Build Packet Print Packet

Generated: 2026-05-02

Packet folder: ` /sessions/charming-bold-fermat/mnt/GitHub/ocarina`

File Map

File	Purpose
`design.md`	Project intent, catalog metadata, assumptions, and validation plan.
`bom.csv`	Starter bill of materials with part categories, quantities, drawing refs, and notes.
`sourcing.csv`	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
`cut-list.csv`	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
`drawing-brief.md`	Manufacturing drawing and technical product sketch brief.
`assembly-manual.md`	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
`validation.csv`	Target/measured values, tolerance, environment, result, and tuning/build action log.
`supplier-rfq.md`	Supplier email/request-for-quote starter.
`visual-bom-brief.md`	Art direction for an image-forward visual BOM.
`wolfram-starter.wl`	Wolfram starter for physics, optimization, visualization, and validation.
`README.md`	Project artifact.

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design.md

Project intent, catalog metadata, assumptions, and validation plan.

Ocarina Build Packet

Source

- Design table: `docs/Ocarina.xlsx`
- Sheet: `Ocarina`
- Inspected range: `A1:H141`
- Workbook content observed: design inputs, size library, Helmholtz calculator, 12-hole fingering chart, ceramic workflow, BOM, design notes, and Wolfram notebook notes.

Design Intent

Build a slip-cast ceramic Alto C 12-hole ocarina using a 3D-printed master and plaster mold workflow. The current table is focused on the standard "sweet potato" vessel-flute form, but the same model can be used as the baseline for sculptural vessel flutes.

Governing Model

Ocarinas behave primarily as Helmholtz resonators:

$$f = c/(2\pi) * \sqrt{A_{\text{open}}/(V_{\text{chamber}} * L_{\text{eff}})}$$

Unlike a transverse flute or Native American style flute, pitch is driven by chamber volume and total open-hole area, not tone-hole distance along a bore. Hole position is still important for ergonomics, grip, and fingering logic.

Current Workbook Inputs

Field	Current value
---	---
Ocarina type	Alto C, 12-hole
Chamber volume	130 cm ³
Wall thickness	0.4 cm
Voicing window width	0.9 cm
Voicing window height	0.5 cm
Speed of sound	34300 cm/s
Clay body	Cone 6 stoneware
Shrinkage	12 percent
Target low note	A4, 440 Hz
Target high note	F6, 1397 Hz

Critical Design Features

- Chamber volume is the master tuning input. Measure it by water fill after CAD, master print, greenware, bisque, and glaze fire.
- Voicing window area sets the all-closed low-note behavior along with volume and effective neck length.
- Fipple geometry is the playability bottleneck: windway gap, labium angle, channel smoothness, and edge sharpness should be treated as first-class dimensions.
- Finger holes should start undersized. Enlarge to sharpen; flattening after over-enlarging is possible but slower and less clean.
- Exterior-only glazing is preferred for tuning stability and acoustic response. Mask the fipple, holes, and chamber interior.

Workbook Improvement Notes

During inspection, rows 39-42 of the 12-hole fingering chart contain formulas for hole area, cumulative open area, and calculated frequency. Rows 43-51 currently show placeholder `=` cells instead of filled formulas. Recommended next workbook cleanup:

1. Copy the area formula pattern from `D40:D42` through `D51`.
2. Copy the cumulative area pattern from `E40:E42` through `E51`.
3. Copy the frequency formula pattern from `F40:F42` through `F51`.
4. Add measured frequency, cents error, and tuning action columns.
5. Add a note that hole combinations can be cross-fingered if total open area is equivalent.
6. Verify the size-library `Length (in)` column. Inspection showed serial-like values such as `46180`, so those cells may have been formatted or entered incorrectly.

Mold Strategy

Use a 2-piece plaster mold for the main body. The first prototype can use a press-molded body to speed iteration, but production should move toward slip casting once the fipple and chamber volume are stable.

Recommended mold/CAD features:

- Split line around the widest body perimeter.
- Positive registration keys that do not trap clay.
- Pour/drain path or assembly opening appropriate to the chosen casting workflow.
- Removable or cleanable fipple detail. If the fipple is too delicate for the mold, cast a blank and hand-cut the voicing at leather-hard stage.
- Master scale factor: $1/(1 - \text{shrinkage})$; for 12 percent shrinkage this is about `1.136`.

Prototype Ladder

| Prototype | Goal | Success criteria |

| --- | --- | --- |

| OCA-P0 voicing tile | Practice windway/labium only | Clear tone on a disposable test cavity |

| OCA-P1 closed vessel | Verify chamber volume and fipple | Stable A4-ish all-closed tone before finger holes |

| OCA-P2 4-hole | Prove tuning workflow | Four tuned notes within +/-25 cents after bisque |

| OCA-P3 12-hole | Full range test | Chromatic pattern playable, no blocked grip |

| OCA-P4 matched set | Soprano C, Alto C, Bass C | Shared mold logic and glaze family |

Open Assumptions

- Pricing in the workbook is an estimate and has not been date-checked.
- Clay shrinkage must be measured for Tony's actual slip, not assumed from supplier average.
- The fipple may need empirical iteration beyond the Helmholtz model.
- Exact ergonomics depend on hand span and final sculptural body shape.

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bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

| item_id | category | item | qty | spec | make_buy | estimated_cost | source_note | drawing_ref | notes |

| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |

| OCA-BOM-001 | Master | 3D printed body master | 1 set | PLA or resin printed top and bottom master | Make | \$2-5 | Workbook estimate not date-checked | OCA-DRW-001 | Scale master for measured clay shrinkage. |

| OCA-BOM-002 | Mold | #1 pottery plaster | 10 lb | USG #1 pottery plaster or equivalent | Buy | \$15-25 | Workbook estimate not date-checked | OCA-DRW-002 | Enough for several small mold sets depending on cottle size. |

| OCA-BOM-003 | Clay | Cone 6 stoneware slip or clay body | 25 lb | Slip casting body or press-moldable clay | Buy | \$20-35 | Workbook estimate not date-checked | OCA-DRW-003 | Record batch and measured shrinkage. |

| OCA-BOM-004 | Finish | Cone 6 exterior glaze | 3-5 pints | Food-safe or durable ceramic glaze | Buy | \$30-60 | Workbook estimate not date-checked | OCA-DRW-004 | Mask fipple, holes, and interior. |

| OCA-BOM-005 | Tuning | Diamond needle files | 1 set | Assorted profiles | Buy | \$10-20 | Workbook estimate not date-checked | OCA-DRW-005 | Needed after bisque and final fire. |

| OCA-BOM-006 | Tuning | Chromatic tuner or analysis app | 1 | Cent-accurate tuner | Buy | \$15-30 | Workbook estimate not date-checked | OCA-VAL-001 | Use same app/settings across prototypes. |

| OCA-BOM-007 | Mold | Mold soap or release | 1 bottle | Murphy's oil soap or mold release | Buy | \$5-10 | Workbook estimate not date-checked | OCA-DRW-002 | For sealed master and plaster parting surfaces. |

| OCA-BOM-008 | Firing | Bisque and glaze firing | per load | Cone 06 bisque and Cone 6 glaze | Buy | \$20-50/load | Workbook estimate not date-checked | OCA-VAL-002 | Record firing schedule and kiln location. |

| OCA-BOM-009 | Masking | Wax resist | 1 small jar | Brushable wax resist | Buy | TBD | Add supplier/date before purchase | OCA-DRW-004 | Protect tuning-critical holes and fipple. |

| OCA-BOM-010 | Measurement | Graduated syringe or burette | 1 | Water-fill volume measurement | Buy | TBD |
Add supplier/date before purchase | OCA-VAL-003 | Needed for chamber volume validation. |

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sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

| item_id | item | required_spec | search_terms | supplier_candidates | date_checked | unit_price | lead_time |
substitution_rule | risk_note |

| --- | --- | --- | --- | --- | --- | --- | --- |

| OCA-SRC-001 | #1 pottery plaster | Pottery mold plaster suitable for absorbent ceramic molds | USG #1 pottery
plaster 10 lb 25 lb | TBD | | | | Use equivalent pottery plaster if absorption and strength are similar | Wrong plaster
can give weak molds or poor casting absorption. |

| OCA-SRC-002 | Cone 6 casting slip | Stoneware slip with known shrinkage and firing schedule | cone 6 stoneware
casting slip shrinkage | TBD | | | | Substitute only after measuring shrinkage bars | Unknown shrinkage invalidates
CAD scale factor. |

| OCA-SRC-003 | PLA or resin | Print material that can be sanded and sealed | PLA filament smooth master mold
making resin print | TBD | | | | Any stable printable material is acceptable if sealed | Layer lines transfer into mold and
cast. |

| OCA-SRC-004 | Mold release | Mold soap compatible with plaster | pottery mold soap plaster release | TBD | | | |
Commercial mold soap preferred | Release failure can damage master or mold. |

| OCA-SRC-005 | Diamond files | Small round and tapered profiles | diamond needle file set ceramic tuning | TBD | | | |
| Diamond burrs can substitute for some work | Coarse tools chip fired ceramic. |

| OCA-SRC-006 | Wax resist | Brushable wax for glaze masking | ceramic wax resist glaze masking | TBD | | | | Latex
or tape only if kiln-safe workflow is confirmed | Glaze in holes or windway can ruin tuning. |

| OCA-SRC-007 | Measurement tools | Cent-accurate tuner plus volume measurement | chromatic tuner cents water
volume graduated syringe | TBD | | | | Phone app acceptable for early tests | Inconsistent measurement hides
process drift. |

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cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

| cut_id | part | qty | rough_dimensions_in | final_dimensions_in | material | orientation | operation | tolerance_in |
yield_or_offcut | notes |

| --- | --- | --- | --- | --- | --- | --- | --- | --- |

| OCA-CUT-001 | Master body upper half (3D-printed) | 1 | 4.50 x 3.50 x 2.50 envelope | master scale =
1/(1-shrinkage); for 12% shrinkage = 1.136x fired body | PLA or sealable resin | X along mouthpiece axis; Z up; mark
split-line and chamber datum | Print, sand layer lines, fill voids, seal/sand, sign with build ID | +/-0.010 | Save offcut
sprues for shrinkage coupons | Master scale factor must be re-derived from MEASURED clay shrinkage before final

master. |

| OCA-CUT-002 | Master body lower half (3D-printed) | 1 | 4.50 x 3.50 x 2.50 envelope | mirror of OCA-CUT-001 | PLA or sealable resin | X along mouthpiece axis; Z up; mark split-line | Print, sand, fill, seal | +/-0.010 | Reuse offcut sprues for shrinkage coupons | Same scale factor as OCA-CUT-001. |

| OCA-CUT-003 | Wooden cottle boards (mold making) | 4 | 8 x 6 x 0.75 each | 7.50 x 5.50 x 0.75 each | Melamine-faced MDF or sealed plywood | Smooth face inward; mark inner clamping zone | Cut, edge-seal, drill clamping holes | +/-0.030 | One sheet of 24 x 18 x 0.75 yields all 4 | Clean parting wax/release before each pour. |

| OCA-CUT-004 | Plaster mother mold half (poured) | 2 | 7.50 x 5.50 x 2.00 each | 7.50 x 5.50 x 1.50-1.75 each (around master) | USG #1 pottery plaster (or equiv.) | Pour with master keyed to first half; second half over registration | Mix to 73:100 water:plaster (by weight); 2-min slake; pour around sealed master | +/-0.060 | Trim spew/edge after demold; recover small offcuts as test slabs | Min 1.50 in plaster around master where practical for stiffness. |

| OCA-CUT-005 | Greenware vessel (slip-cast) | 1 per pour | ~3.95 x 3.07 x 2.20 (master cavity wet) | ~3.50 x 2.72 x 1.95 fired (12% shrink) | Cone 6 stoneware casting slip | Mouthpiece axis along master split | Slip pour > drain at target wall (~0.16 in / 0.4 cm) > demold leather-hard | N/A (process-controlled) | Reuse demold trim/fettling waste as slip recycle | Wall thickness drives drying time and tap tone — measure on noncritical trim. |

| OCA-CUT-006 | Voicing window cut (leather-hard) | 1 | N/A | 0.354 x 0.197 (window W x H = 0.9 x 0.5 cm per design) | Cone 6 greenware | Cut perpendicular to mouthpiece axis | Score, lift, clean labium edge with rib | +/-0.010 | N/A | Cut undersized; tune by enlarging at bisque tuning step. |

| OCA-CUT-007 | Finger holes (12-hole layout — leather-hard) | 12 | N/A | 0.16-0.31 dia each (start undersized; see design table OCA-DRW-005) | Cone 6 greenware | Per fingering chart row D40:F51 of design table | Brass tube cutter or hole punch; clean burr with damp finger | +/-0.010 (start) | N/A | All 12 holes start at design-table 'minimum' diameter; enlarge at tuning. |

| OCA-CUT-008 | Post-bisque tuning trim | as needed | N/A | Within +/-25 cents of target frequency per validation.csv | Bisqued ceramic | N/A | Diamond needle file or fine diamond burr in steps | +/-0.005 dimensional; +/-25 cents acoustic | Diamond burr offcuts are kiln dust — bag and discard | Document each enlarge step on validation row. |

| OCA-CUT-009 | Wax resist mask (pre-glaze) | 1 application | N/A | ~0.020 in dry film over voicing window + all 12 holes + interior | Brushable wax resist | N/A | Brush-apply; dry per supplier; check all openings clear | Visual | N/A | Mask all acoustic edges to keep glaze out of windway and holes. |

| OCA-CUT-010 | Glaze application (exterior only) | 1 application | N/A | ~0.005-0.010 in per coat; 1-2 coats exterior | Cone 6 satin or matte glaze | Avoid waxed regions | Dip / brush / spray exterior only | +/-0.005 (visual) | N/A | Glaze in voicing window or holes will detune the instrument. |

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drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Ocarina Drawing Brief

Required Views

- Top view with body outline, hole layout, hand assignment, and centerline.

- Side view with mouthpiece axis, windway height, labium angle, body thickness, and split line.
- Section through fipple showing windway, labium, window, and chamber connection.
- Section through body showing chamber volume reference, target wall thickness, and mold split.
- Detail view for hole diameters and post-fire tuning allowance.
- Mold view showing plaster parting line, registration keys, pour/drain plan, and cottle clearance.

Critical Dimensions

Dimension Source Tolerance intent
--- --- ---
Chamber volume Workbook input and water-fill measurement Tuning critical
Voicing window width/height Workbook input Playability critical
Windway gap CAD/test-piece measurement Playability critical
Labium angle and edge thickness CAD/test-piece measurement Playability critical
Wall thickness Workbook input and cast measurement Tone/drying critical
Hole diameters Workbook chart and tuning log Tuning critical
Master scale factor Measured clay shrinkage Process critical

Notes For CAD

- Store chamber volume as a named parameter.
- Store shrinkage factor as a named parameter.
- Separate final fired dimensions from master dimensions.
- Mark fipple geometry as prototype-sensitive; do not freeze it until test pieces speak reliably.
- Include a versioned body ID so castings can be tied back to the CAD master.

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assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Ocarina Assembly Manual

Scope

This manual covers a slip-cast or press-molded ceramic ocarina based on `docs/Ocarina.xlsx`. It is written for prototype builds where the first goal is repeatable tone and validated tuning, not decorative finish.

Tools

- CAD package for master design.
- 3D printer and sanding/sealing supplies.
- Cottle boards, plaster mixing bucket, scale, mold soap, and clamps.
- Stoneware slip or workable clay body.
- Small hole cutters, drill bits, needle tools, loop tools, ribs, sponge, and fettling knife.
- Chromatic tuner, microphone, calipers, graduated syringe or water-fill vessel.
- Diamond needle files or fine diamond burrs.
- Wax resist, glaze, and kiln access.

Process

1. **Set design inputs**

- Confirm target ocarina size, chamber volume, voicing window, wall thickness, clay body, and shrinkage.
- Create a build ID before CAD begins.

2. **Model the master**

- Model the body as a controlled-volume vessel.
- Include split-line planning, registration features, and a fipple strategy.
- Scale the master by $\frac{1}{1 - \text{measured_shrinkage}}$.

3. **Print and finish the master**

- Print the master with enough wall strength to survive mold making.
- Sand layer lines and seal the surface.
- Mark datums: centerline, mouthpiece axis, split plane, and hole reference side.

4. **Make the mold**

- Apply release to the sealed master and cottle surfaces.
- Pour plaster with at least 1.5 inches around the master where practical.
- Add registration keys and dry the mold fully before casting.

5. **Cast or press the body**

- For slip casting, pour slip, wait for target wall buildup, drain, and demold at leather-hard stage.
- For press molding, press even clay slabs into both mold halves and join with scored slip.
- Measure sample wall thickness at a noncritical trimmed area.

6. **Cut fipple and holes**

- Establish the windway first.

- Cut the labium cleanly and test blow before the body fully dries.
- Cut finger holes undersized.

7. ****Dry slowly****

- Dry under plastic until moisture equalizes.
- Watch the mouthpiece and seam areas for cracks.

8. ****Bisque fire****

- Bisque to the clay body's recommended schedule.
- Measure mass, volume, hole diameters, and frequencies.

9. ****Tune after bisque****

- Enlarge holes to raise pitch.
- Refine the fipple only in small steps.
- Record measured frequency and cents error after each change.

10. ****Glaze****

- Wax resist the windway, labium, holes, and any interior opening.
- Keep glaze away from the acoustic edges.

11. ****Final fire and validation****

- Fire to the chosen glaze schedule.
- Re-measure all tuning points.
- Save final dimensions and tuning data back into the validation log.

Failure Modes To Watch

- Weak or no tone: windway too tall, labium too blunt, air jet missing edge.
- Pitch too flat: chamber too large, open area too small, or neck length too long.
- Pitch too sharp: chamber too small, open area too large, or over-fired holes.
- Warped mouthpiece: uneven drying or thin unsupported fipple geometry.
- Glaze-choked note: glaze reduced hole or windway opening.

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validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

| build_id | stage | date | clay_body | shrinkage_expected_pct | master_scale_factor | chamber_volume_cm3 |
wall_thickness_cm | voicing_w_cm | voicing_h_cm | hole_id | target_note | target_freq_hz | measured_freq_hz |

cents_error	action	result	notes
---	---	---	---
OCA-P0	voicing_tile		all_closed A4 440 Practice fipple before full body.
OCA-P1	greenware		130 0.4 0.9 0.5 all_closed A4 440 Measure before drying if possible.
OCA-P1	bisque		TBD TBD TBD TBD all_closed A4 440 First real model check.
OCA-P1	glaze_fire		TBD TBD TBD TBD all_closed A4 440 Record glaze shift.
OCA-P2	bisque		TBD TBD TBD TBD hole_1 Bb4 466.2 Start undersized and enlarge gradually.
OCA-P2	bisque		TBD TBD TBD TBD hole_2 B4 493.9 Record hole diameter after tuning.
OCA-P2	bisque		TBD TBD TBD TBD hole_3 C5 523.3 Check breath pressure sensitivity.
OCA-P3	glaze_fire		TBD TBD TBD TBD hole_12 A5 880 Full 12-hole validation row.

supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ — Slip-Cast Ceramic Ocarina (OCA-001 family)

> Use this as a template for outreach to ceramic-supply, plaster, mold-making, and finishing suppliers. Customize the Subject and address fields per recipient.

Subject: RFQ — slip-cast ceramic ocarina prototype materials and consumables

Hello,

I'm a mechanical R&D engineer prototyping a small family of slip-cast ceramic ocarinas (vessel flutes, ~3-4 in long, ~0.16 in / 4 mm wall) and need quotes for the following. The first build is an Alto C 12-hole prototype based on the parametric Helmholtz design table at [tonykoop/ocarina](https://github.com/tonykoop/ocarina).

Items

#	Item	Required spec	Approx. qty (per prototype run)
1	Cone 6 stoneware casting slip	Mature Cone 6 stoneware slip with published shrinkage (X/Y/Z), water absorption % , and firing schedule . White or light buff body preferred for tone consistency over dark iron-bearing slips.	1-5 gal
2	#1 pottery plaster	USG #1 Pottery Plaster (or equivalent) suitable for absorbent slip-cast molds. Need water:plaster ratio , set time , soluble salt content .	10 lb (single mold) — 50 lb (production mold set)

| 3 | Mold release / mold soap | Plaster-compatible parting agent (Murphy's-type oil soap or commercial mold soap). | 1 bottle / 16 oz |

| 4 | Cone 6 exterior glaze | Food-safe (or durable non-food-safe) satin or matte glaze. Color: TBD per family — likely earth/buff/cream. Glaze must not bridge holes or windway. | 3-5 pints |

| 5 | Brushable wax resist | Standard ceramic wax resist for masking voicing window, finger holes, and chamber interior pre-glaze. | 1 small jar / 8 oz |

| 6 | Diamond needle file set (post-bisque tuning) | Assorted profiles (round, half-round, taper) ~3-6 mm; resin-bonded acceptable. | 1 set |

| 7 | Hole cutters / brass tube cutters | Round cutters from ~0.156 in (4 mm) to ~0.313 in (8 mm) diameter for greenware finger-hole cutting. | 1 set |

| 8 | Bisque + glaze firing service (if local) | Cone 06 bisque + Cone 6 glaze fire; small cubic-foot kiln load. | 2 fires per prototype |

| 9 | (Optional) Master printing/outsource | If you do high-resolution PLA or resin prints up to ~5 in long with **0.005 in dimensional accuracy**, please quote. We currently print in-house but want a backup. | 1-5 master pairs |

What we need in your quote

- **Unit price** and any volume breaks
- **Minimum order quantity**
- **Current stock status** and **lead time**
- **Shipping estimate** to 94566 (Pleasanton, CA) — or local pickup option
- **Material safety data sheet (MSDS)** and **technical data sheet** for items 1, 2, 4, 5
- **Recommended substitutions** for thin-walled (~4 mm) musical instrument bodies
- For item 1: most recent **shrinkage test data** for your batch (we will re-measure on our side, but a baseline matters)

Acceptance criteria

The prototypes are measured for X/Y/Z shrinkage, bore/cavity volume after firing, wall-thickness consistency, and acoustic tuning (chromatic tuner, ± 25 cents target post-bisque). Slip with unknown or wildly variable shrinkage invalidates the master scale factor (currently $1/(1-0.12) = 1.136$), so reproducibility data is more valuable than headline price.

The repo is a private R&D portfolio repo; we do not need any IP-restricted documents. Public-domain MSDS and supplier spec sheets are sufficient.

Thank you,

Tony Koop

Mechanical R&D Engineer

tonykoop@gmail.com

github.com/tonykoop/ocarina (private; access on request)

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visual-bom-brief.md

Art direction for an image-forward visual BOM.

Ocarina Visual BOM Brief

Goal

Create a one-page visual BOM that helps a builder understand the ceramic ocarina workflow from digital master to tuned fired instrument.

Layout

- Header: "Slip-Cast Ceramic Ocarina - Alto C Prototype".
- Hero image: finished ocarina or clean CAD render.
- Process strip: CAD master, 3D print, plaster mold, greenware, bisque, tuning, glaze fire.
- BOM table: item number, part/material, quantity, spec, estimated cost, make/buy, image.
- Callout inset: fipple section with windway, labium, and voicing window labels.
- Validation inset: chamber volume, all-closed note, highest tested note, cents error.

Image Requirements

Use real shop photos when available. Generated or rendered placeholders should be labeled as placeholders until replaced with:

- Master print photo.
- Plaster mold photo.
- Greenware body photo.
- Fipple close-up.
- Final fired instrument.
- Tuning tool/photo of measurement setup.

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wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

(* Ocarina Helmholtz notebook starter *)

```
ClearAll["Global`*"];
```

```
c = 34300; (* cm/s at about 20 C *)
```

```
targetA4 = 440;
```

```
volumeCm3 = 130;
```

```
wallCm = 0.4;
```

```
windowWcm = 0.9;
```

```
windowHcm = 0.5;
```

```
voicingAreaCm2[w_, h_] := w*h;
```

```
endCorrectionCm[area_] := 0.6*Sqrt[area/Pi];
```

```
helmholtzHz[area_, volume_, neckLength_] :=
```

```
(c/(2*Pi))*Sqrt[area/(volume*neckLength)];
```

```
centsError[measured_, target_] := 1200*Log[2, measured/target];
```

```
baseArea = voicingAreaCm2>windowWcm, windowHcm];
```

```
baseNeck = wallCm + endCorrectionCm[baseArea];
```

```
baseFreq = helmholtzHz[baseArea, volumeCm3, baseNeck];
```

```
holeDiameterMmToAreaCm2[d_] := Pi*(d/20)^2;
```

```
targetFreq[ midi_] := 440*2^((midi - 69)/12);
```

```
areaForFrequency[f_, volume_, neckLength_] :=
```

```
volume*neckLength*(2*Pi*f/c)^2;
```

(* Candidate cumulative area table for A4 to A5. *)

```
notes = {
```

```
{ "A4", 69}, {"Bb4", 70}, {"B4", 71}, {"C5", 72},
```

```
{ "C#5", 73}, {"D5", 74}, {"D#5", 75}, {"E5", 76},
```

```
{ "F5", 77}, {"F#5", 78}, {"G5", 79}, {"G#5", 80}, {"A5", 81}
```

```
};
```

```
areaTable = Table[
```

```
With[{freq = targetFreq[ midi]},
```

```
{name, freq, areaForFrequency[freq, volumeCm3, baseNeck]}
```

```
],
```

```
{ {name, midi}, notes}
```

```
];
```

```
areaTable // TableForm
```

```
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```

README.md

Project artifact.

Ocarina

Engineering documentation and parametric design table for the ocarina — the vessel flute family in ceramic, wood, and 3D-printed forms.

Part of the [tonykoop/instrument-maker](https://github.com/tonykoop/instrument-maker) catalogue.

> CAD renders, Wolfram notebook recordings, and a finalized build method are forthcoming.

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